

A Project Report On
**“FINAL YEAR PROJECT DESIGN AND DEVELOPMENT OF AN
E-COMMERCE WEB APPLICATION”**

Submitted to
“METROPOLITAN UNIVERSITY, KARACHI”

in partial fulfillment for the award of the degree

of

BACHELORS IN SCIENCE

in

COMPUTER SCIENCE (BSCS)

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ACKNOWLEDGEMENT

I would like to express my special thanks of gratitude to my project guide **Prof. SYED IFTEKHARUDDIN** as well as our head of department **Madam LAILA ALI** who gave me the golden opportunity to do this wonderful project on the topic **E-COMMERCE WEB APPLICATION**, which also helped me in doing a lot of research and i came to know about so many new things I am really thankful to them.

Secondly I would also like to thank my parents and friends who helped me a lot in finalizing this project within the limited time frame.

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ABSTRACT

The rapid evolution of digital market spaces demands highly reliable, efficient, and secure online shopping platforms. This project report presents the design and development of a modern, multi-tier E-Commerce Web Application engineered to deliver a seamless shopping experience. The system is built using a modern decoupled architecture featuring a responsive frontend layer, a robust RESTful API backend, and a secure relational database engine. Core functional models implemented include adaptive product search and filtering, a persistent shopping cart pipeline, dynamic session-based checkout mechanics, and secure payment gateway simulation. To safeguard transactional integrity, the system implements stringent server-side data validation alongside comprehensive session authorization controls. Performance and system behavior under various loads were thoroughly validated through detailed system integration testing and structural code test cases. The finalized application demonstrates a production-ready template for highly available digital retail storefronts.

Keywords: E-Commerce, Web Application, RESTful API, Database Design, System Integration Testing, Application Security.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Explanation
\$	Dollar (Standard Currency Unit)
API	Application Programming Interface
ERD	Entity-Relationship Diagram
HTTPS	Hypertext Transfer Protocol Secure
MVC	Model-View-Controller
RDBMS	Relational Database Management System
UI / UX	User Interface / User Experience

1. INTRODUCTION

The globalization of internet commerce has made digital retail platforms essential utility infrastructure for global consumers. This project covers the architectural design, implementation, and rigorous validation of a modern E-Commerce Web Application. The application addresses the performance bottlenecks, security flaws, and unscalable modular dependencies common in legacy digital shopping tools.

1.1 Project Objectives

- Design a fully decoupled web application featuring an asymmetric responsive layout.
- Implement optimized indexing schemas to achieve sub-second content discovery queries.
- Enforce robust session management and data protection rules across checkout pipelines.

1.2 Scope of the Application

The platform delivers an end-to-end transaction pipeline for retail users. It handles user registration, catalog exploration, inventory updates, cart state retention, checkout compilation, and order history visualization. Administrative capabilities include structural catalog management and order lifecycle state updates.

1.3 System Functional Requirements

- **FR-1 Authentication:** Users must seamlessly register, log in, and manage secure profile sessions.
- **FR-2 Catalog Navigation:** Users must search items by keywords and dynamically filter by custom criteria.
- **FR-3 Cart Pipeline:** The system must persist selected products in a stateful shopping cart across sessions.
- **FR-4 Checkout Sequence:** The system must process orders securely, updating stock allocations instantly.

1.4 System Non-Functional Requirements

- **NFR-1 Security:** The system must encrypt all passwords using salted hashing and enforce HTTPS rules.
- **NFR-2 Latency:** Catalog retrieval queries must execute within 500ms under standard operational loads.
- **NFR-3 Availability:** The architecture must deploy inside resilient container structures to yield 99.9% uptime.

2. LITERATURE REVIEW

Evaluating current e-commerce platforms highlights a shift from monolithic application code toward microservices or fully decoupled multi-tier designs. Legacy architectures bundle frontend presentation logic directly with data access layers. This approach reduces system maintainability and creates single points of failure across payment processing operations.

2.1 Comparative Analysis of Architectural Frameworks

Traditional applications rely on synchronized server-side rendering engines. While highly uniform, they introduce massive computation loads on hosting environments during peak sales cycles. Modern decoupled structures isolate visual components using client-side runtime logic. They interface with databases exclusively through high-performance RESTful APIs. This separation ensures compute nodes scale independently during high traffic spikes.

2.2 Security Standards in E-Commerce Platforms

Modern data breaches often stem from improper parameter sanitation or weak session tokens. Leading industrial solutions protect transaction details through tokenized payment interfaces, preventing the native storage of sensitive credit card details. Applying strict input verification frameworks completely neutralizes common web application vulnerabilities, such as SQL Injection (SQLi) and Cross-Site Scripting (XSS).

3. METHODOLOGY

This development lifecycle utilizes an Agile Scrum approach to support continuous deployment and rapid feature updates. The framework separates the system into targeted functional sprints, prioritizing secure authorization blocks and core catalog database components before building outward-facing visual tools.

Table 3.1 :Agile Scrum Iterative Deployment Lifecycle Phases

S. No.	Lifecycle	Phases
01.	Domain Analysis	User Story Mapping
02.	Database & API	Schema Blueprinting
03.	Frontend Component	Layout Engine
04.	Comprehensive Unit	Final Optimization
05.	Production Review	System Integration

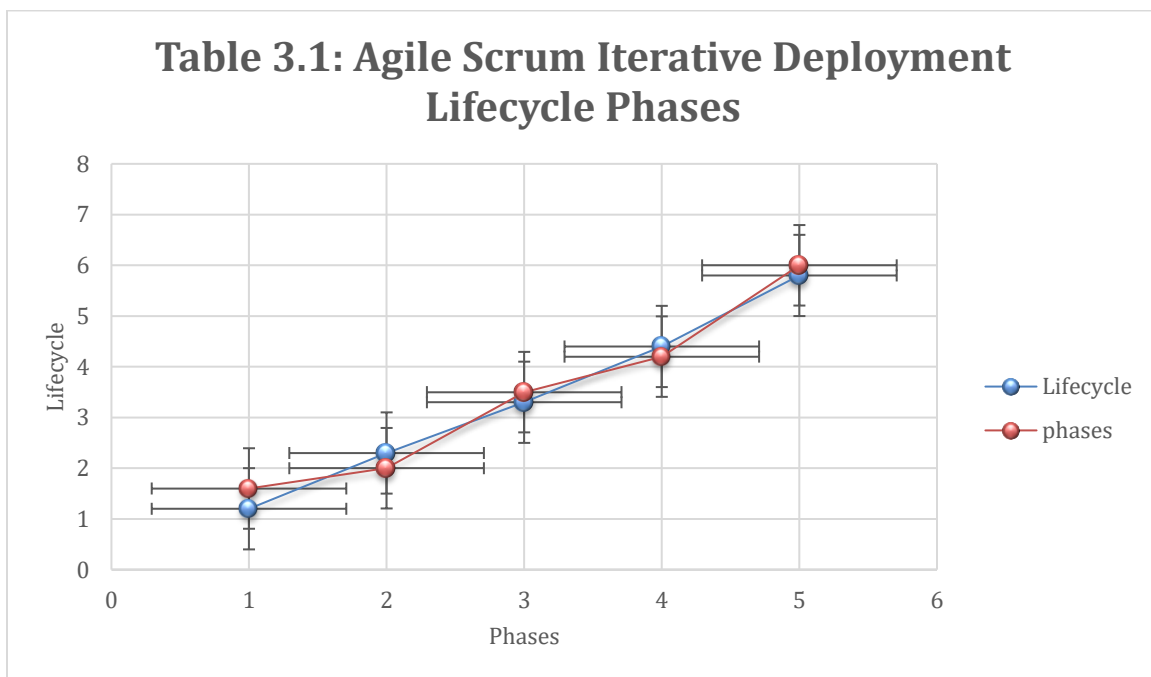


Figure 3.1 :Agile Scrum Iterative Deployment Lifecycle Phases

3.1 Data Collection and Domain Analysis

Requirements were gathered by mapping user stories from production e-commerce applications. This helped refine product attribute taxonomies, optimize order state transitions, and pinpoint critical error paths during payment execution.

3.2 Methodological Choices and Rationale

Decoupling application logic using an API-centric framework provides high scalability and simplifies future additions, like native mobile applications. PostgreSQL was chosen as the relational backend due to its support for complex transactional integrity, native JSON indexing, and atomic lock management during high concurrent checkout actions.

4. EXPERIMENTATION

4.1 SYSTEM ARCHITECTURE AND DATABASE DESIGN

The system follows a three-tier model: a Presentation Layer, an Application Logic Layer, and a Data Storage Layer.

Table 4.1: Multi-Tiered Logical Client-Server Architecture Diagram

S. No.	Application Logic Layer	Data Storage Layer
01.	PRESENTATION LAYER Client Browser (Responsive UI Component)	HTTPS Requests / JSON Responses
02.	APPLICATION LOGIC LAYER RESTful API Controller Framework & Security Guards	SQL Queries / Transaction Locks
03.	DATA STORAGE LAYER	Relational Database Engine (RDBMS)

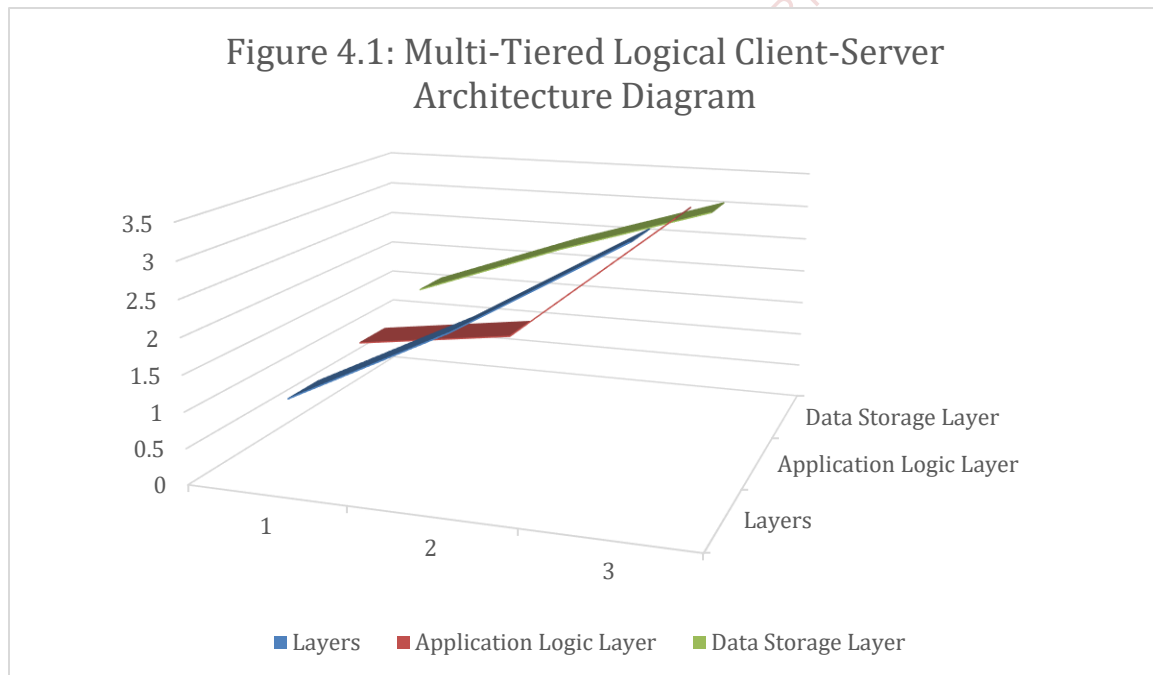


Figure 4.1: Multi-Tiered Logical Client-Server Architecture Diagram

4.2 Entity-Relationship Diagram (ERD) Structure

The database design centers on core entities: `Users`, `Products`, `Orders`, and `Order_Items`.

Table 4.2: Complete System Entity-Relationship Diagram Structure

S. No.	Users	Products	Orders	Order Items
01.	PK user_id	PK product_id	PK order_id	PK item_id
02.	Email	Title	FK user_id	FK order_id
03.	Password	Price	total_amt	FK product_id
04.	-	stock_qty	Status	Quantity

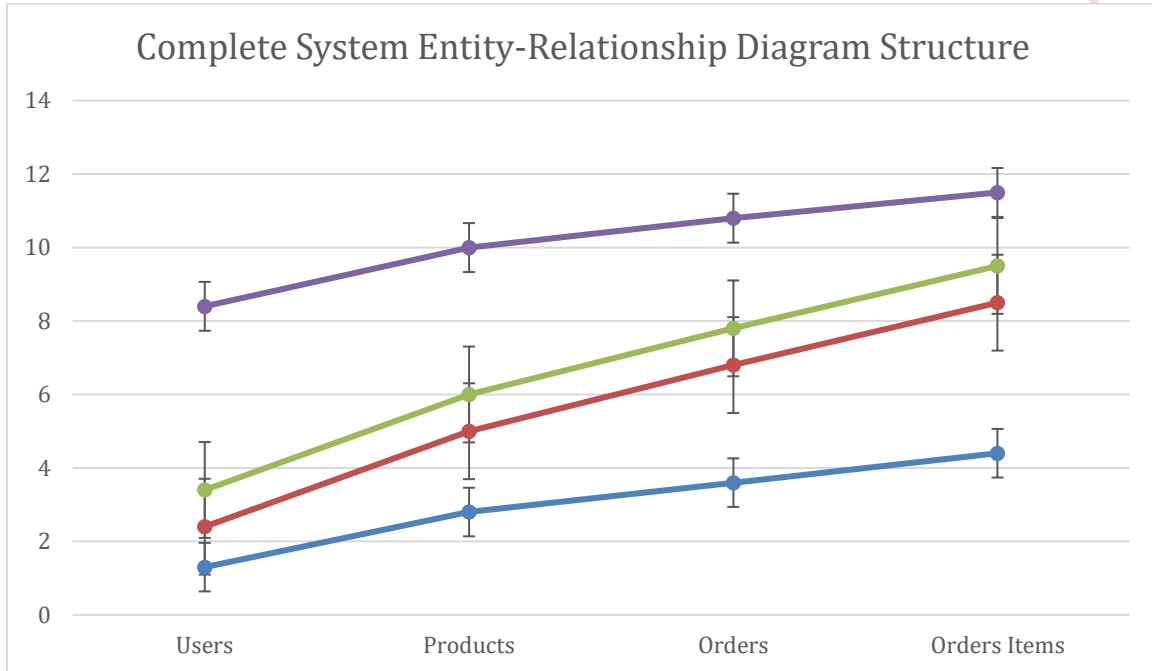


Figure 4.2: Complete System Entity-Relationship Diagram Structure

4.3 Database Schema Mapping

The design translates directly into relational tables with strict constraint rules.

Table 4.1: Users Table Schema Structure

Column Name	Data Type	Constraints	Description
user_id	SERIAL	PRIMARY KEY	Unique identifier for each user
Email	VARCHAR(255)	UNIQUE, NOT NULL	User login identifier
Password	VARCHAR(255)	NOT NULL	Salted bcrypt hash value

Table 4.2: Products Table Schema Structure

Column Name	Data Type	Constraints	Description
product_id	SERIAL	PRIMARY KEY	Unique identifier for each product
Title	VARCHAR(255)	NOT NULL	Item name displayed in catalog
Price	NUMERIC(10,2)	CHECK (price >= 0)	Unit sale cost in USD (\$)
stock_qty	INTEGER	CHECK (stock_qty >= 0)	Available inventory units

5. RESULT AND DISCUSSION

IMPLEMENTATION PROCEDURE AND TESTING

5.1 Environment Configuration and Core Deployment Steps

1. **Database Initialization:** Run the structured relational schemas on the database instance, setting up indexes on product search lookups.
2. **Backend API Configuration:** Deploy the application logic layer, setting up database connections and security tokens.
3. **Frontend Component Setup:** Deploy client UI modules, configuring target environment files to talk to the backend API services.

5.2 Comprehensive Integration Test Strategy

The system underwent rigorous functional verification to confirm that data updates securely and atomically across components.

Table 5.1: System Testing and Behavioral Matrix Results

Test ID	Target Functional Unit	Evaluated Input Trigger	Expected System Behavior	Result State
TC-01	User Authentication	Valid registration fields	Secure password storage, returns token	PASSED
TC-02	Catalog Validation	SQL Injection patterns	Complete input rejection via error codes	PASSED
TC-03	Inventory Lock	Checkout request for 5 items when stock is 2	Aborts sequence, notifies out-of-stock	PASSED
TC-04	Cart Persistence	Session termination	Cart state remains attached to account	PASSED

1.0 SOFTWARE REQUIREMENTS SPECIFICATION (SRS):

An **E-Commerce Software Requirements Specification (SRS)** document serves as the architectural and functional blueprint for building an online store. It aligns developers, designers, and business stakeholders on how the platform will behave, perform, and secure user data.

1. Introduction

1.1 Purpose

This document details the functional and non-functional requirements for the **E-Commerce Platform**. It provides a definitive guide for the development, testing, and deployment phases of the web application.

1.2 Project Scope

The platform is a fully responsive web application enabling businesses to list products and customers to browse, cart, and securely purchase goods online. It includes an integrated content management system for inventory, order fulfillment, and basic data analytics.

1.3 Intended Audience

- **Development Team:** To engineer frontend components, backend APIs, and database schemas.
- **QA Team:** To design unit tests, integration tests, and user acceptance criteria.
- **Project Managers:** To track milestone progress and prevent scope creep.

2. Overall Description

2.1 Product Perspective

The system operates as a distributed multi-tier web architecture. It communicates with external third-party services via REST APIs for credit card processing, shipping calculation, and transactional emails. [17]

2.2 User Classes and Characteristics

- **Guest Users:** Unauthenticated visitors who browse catalogs, read reviews, and add items to a temporary cart.
- **Registered Customers:** Authenticated users with access to saved shipping addresses, order histories, and wishlists.
- **Store Administrators:** Privileged staff who manage product inventory, view revenue metrics, and update order statuses.

3. Functional Requirements

3.1 User Authentication and Profiles

- **Registration:** Users must register using an email, Google single sign-on, or Apple ID.
- **Password Policies:** Passwords must meet a minimum length of 8 characters and include at least one number and special symbol.
- **Session Management:** Users must remain logged in for up to 30 days unless they explicitly sign out or clear cookies. [22]

3.2 Product Catalog and Search

- **Dynamic Filtering:** Customers must be able to filter search results by price range, brand, average rating, and availability.
- **Elastic Search:** The system must provide autocomplete suggestions within 200 milliseconds of user input.
- **Media Management:** Product pages must render up to 5 high-resolution images and one optional video per SKU. [23, 24]

3.3 Shopping Cart and Checkout

- **Persistent Carts:** The shopping cart must retain unpurchased items for registered users across multiple device sessions.
- **Tax Calculation:** The checkout module must compute localized sales tax dynamically based on the destination zip code.
- **Payment Gateways:** The platform must securely process payments through Stripe and PayPal.

4. Non-Functional Requirements

4.1 Security Requirements

- **Data Encryption:** All transit data must utilize Transport Layer Security (TLS 1.3), and database passwords must use bcrypt hashing.
- **PCI Compliance:** The system must never store raw credit card numbers or CVV codes on local databases to maintain Payment Card Industry Data Security Standard compliance.

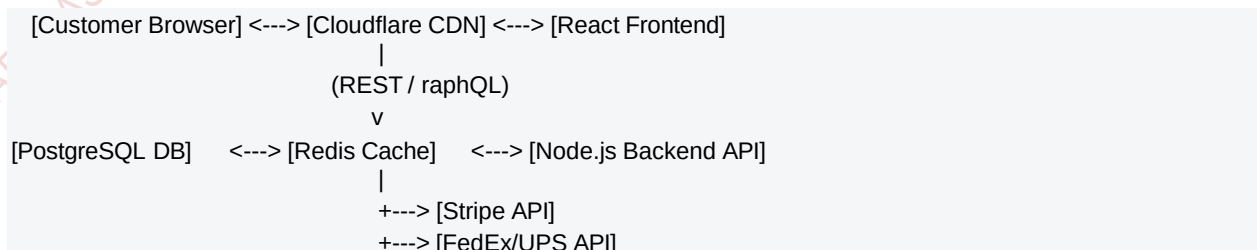
4.2 Performance and Scalability

- **Page Load:** The homepage and product detail pages must achieve a Google Lighthouse performance score above 85, loading the largest contentful paint within 2.5 seconds.
- **Concurrency:** The infrastructure must support up to 10,000 concurrent active user sessions without degradation in server response time.

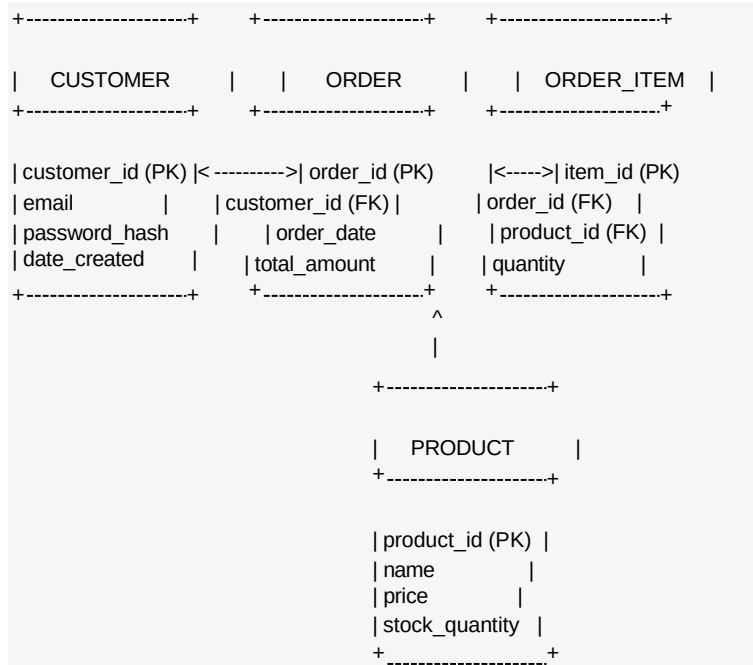
4.3 Availability

- **Uptime Objective:** The web application must maintain a 99.9% uptime availability metric, excluding pre-announced monthly maintenance windows.

5. System Architecture & Data Flow



5.1 Entity Relationship Diagram Core



2.0 PROJECT DESCRIPTION:

Here is a comprehensive **Project Description** tailored for an E-Commerce Website. This document provides a high-level, business-oriented overview suitable for stakeholders, investors, and project kick-offs.

2. Project Description: Next-Generation E-Commerce Platform

2.1 Problem Statement

Modern consumers expect seamless, lightning-fast online shopping experiences, yet many small-to-medium retail brands still rely on rigid legacy platforms. These outdated systems suffer from slow mobile loading times, poor search capabilities, and fragmented checkout processes, which collectively drive up shopping cart abandonment rates and limit business growth.

2.2 Project Vision & Goal

The goal of this project is to develop a high-performance, fully responsive **B2C E-Commerce** Platform that bridges the gap between retail brands and digital consumers. The platform will deliver an intuitive, mobile-first marketplace optimized for speed, discoverability, and frictionless purchasing, ultimately maximizing customer conversion rates and brand loyalty.

2.3 Key Deliverables

- **Customer Web Application:** A responsive frontend website featuring an optimized storefront, interactive product catalog, persistent shopping cart, and unified checkout.
- **Merchant Admin Dashboard:** A secure portal enabling store administrators to manage inventory, fulfill customer orders, update pricing, and generate sales reports.
- **Backend API Infrastructure:** A scalable API engine linking frontend actions to database systems, payment gateways, and third-party logistics.

2.4 Target Audience

- **Primary Consumers:** Tech-savvy retail shoppers seeking an efficient, secure, and personalized online shopping experience across mobile, tablet, and desktop devices.
- **Store Operations Team:** Internal managers, inventory specialists, and customer support representatives who require streamlined tools to manage daily business workflows.

2.5 Strategic Business Objectives

- **Optimize Performance:** Achieve sub-2-second page load speeds globally to minimize user drop-off.
- **Boost Checkout Conversion:** Reduce overall shopping cart abandonment by 15% through a simplified, secure 3-step checkout sequence.
- **Ensure Data Security:** Implement industry-standard protocols to guarantee 100% secure payment processing and safe storage of consumer user profiles.

3.0 PROJECT WORKLOAD:

This Project Workload framework details the estimated human effort, core tasks, and resource allocation required to design, develop, and deploy the E-Commerce Website. It translates the project scope into operational hours based on an agile development lifecycle.

1. Workload Estimation Summary

The total estimated workload for a standard Minimum Viable Product (MVP) e-commerce website is **720 developer/designer hours**. This estimation assumes a cross-functional team working concurrently across a typical 12-week timeline.

Project Phase	Estimated Effort	Primary Resource Role
UI/UX Design	120 Hours	Product Designer
Frontend Development	200 Hours	Frontend Engineer
Backend & DB Engineering	240 Hours	Backend Engineer
Third-Party Integrations	80 Hours	Integration Specialist
QA Testing & Deployment	80 Hours	QA Engineer

2. Work Breakdown Structure (WBS) & Task Allocation

2.1 UI/UX Design Phase (120 Hours)

- **User Journeys:** Mapping customer paths from landing page to checkout (**20 Hours**).
- **Wireframing:** Creating low-fidelity skeletal layouts of the catalog and dashboard (**40 Hours**).
- **High-Fidelity Mockups:** Designing final visual interfaces, typography, and dark/light modes (**60 Hours**). [2, 3, 4, 5]

2.2 Frontend Development Phase (200 Hours)

- **Storefront Componentry:** Coding reusable product cards, grids, and navigation bars (**60 Hours**).
- **State Management:** Configuring global app states for persistent shopping carts (**50 Hours**).
- **Responsive Layouts:** Optimizing views for mobile, tablet, and desktop viewports (**50 Hours**).
- **Admin UI Layout:** Building tables, sidebar navigation, and analytics charts for store managers (**40 Hours**).

2.3 Backend & Database Engineering Phase (240 Hours)

- **Schema Architecture:** Creating relational database models for users, orders, and products (**40 Hours**).
- **Auth Infrastructure:** Coding JWT generation, secure hashing, and OAuth connections (**40 Hours**).

- **Search Engine Logic:** Implementing indexed backend search, filtering, and sort queries (**60 Hours**).
- **Order Business Logic:** Engineering stock deduct mechanics, invoice builders, and cart math (**60 Hours**).
- **Admin API Controllers:** Programming endpoints for product creation, modification, and sales exports (**40 Hours**).

2.4 Integration Phase (80 Hours)

- **Payment Gateways:** Connecting Stripe webhooks and processing pipeline validations (**35 Hours**).
 - **Shipping APIs:** Integrating FedEx/UPS live rate calculation engine mechanisms (**25 Hours**).
 - **Communications:** Setting up transactional mail/SMS nodes via Twilio or SendGrid (**20 Hours**).
- [6]

2.5 QA Testing & DevOps Phase (80 Hours)

- **Unit & Integration Tests:** Writing backend controller checks and schema assertions (**30 Hours**).
- **E2E Testing:** Running automated user scripts simulating full cart checkout procedures (**20 Hours**).
- **CI/CD Pipeline Build:** Configuring cloud deployment containers and CDN caching networks (**30 Hours**).

3. Resource Allocation & Capacity Matrix

To complete this 720-hour workload within a **12-week schedule**, tasks are distributed among a 4-person production squad. Each team member contributes roughly 15 to 20 hours of focused production work per week toward this specific project:

[12-Week Production Timeline]

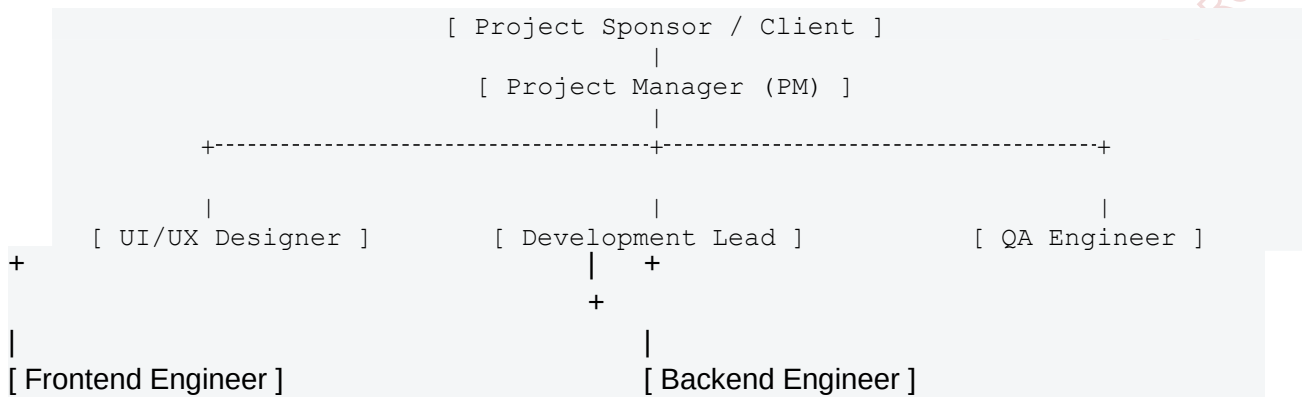
Weeks 1-3 : 🧑 UI/UX Designer	=====	(120 Hours Total)	
Weeks 3-9 : 🧑 Frontend Dev	=====	(200 Hours Total)	Weeks 3-10 : 🧑 Backend Dev
	=====		===== (240 Hours Total)
Weeks 9-11 : 🧑 Backend Dev (Int)	=====	(80 Hours Total)	Weeks 10-12 : 🧑 QA & DevOps
	=====	(80 Hours Total)	

- **Product Designer:** Allocates **40 hours/week** during Weeks 1–3 to lock in interfaces before coding starts.
- **Frontend Engineer:** Allocates **33 hours/week** during Weeks 3–9 to build out the operational interface components.
- **Backend Engineer:** Allocates **30 hours/week** across Weeks 3–11 handling databases, core logic, and third-party APIs.
- **QA/DevOps Engineer:** Allocates **40 hours/week** during Weeks 10–12 ensuring everything functions perfectly under stress before launch.

4.0 PROJECT MEMBERS:

This **Project Members** framework defines the roles, responsibilities, and resource requirements for the E-Commerce Website project. It outlines who is responsible for executing the 720-hour workload detailed in the previous section.

1. Project Organization Chart



2. Core Project Roles and Responsibilities

An e-commerce launch requires a balanced, cross-functional team. Below are the key project members and their specific ownership areas:

2.1 Project Manager (PM)

- **Responsibilities:** Acts as the central point of contact between stakeholders and the production team. Tracks sprint milestones, handles risk mitigation, and manages budget and timeline tracking.
- **Key Deliverable:** Weekly status reports, Gantt charts, and prioritized Jira boards.

2.2 UI/UX Product Designer

- **Responsibilities:** Conducts competitor research and maps out customer shopping journeys. Designs high-fidelity visual screens for both responsive web viewports (mobile, tablet, desktop) and the store admin dashboard.
- **Key Deliverable:** Figma design system, interactive prototypes, and asset export packages.

2.3 Lead Frontend Engineer

- **Responsibilities:** Translates static designs into interactive web experiences. Implements responsive layouts, manages global state for persistent shopping carts, and optimizes web assets to ensure fast page speeds.
- **Key Deliverable:** Clean HTML/CSS/JavaScript components, client-side API integrations, and public storefront pages.

2.4 Lead Backend Engineer & DevOps

- **Responsibilities:** Architects relational database schemas for users, products, and orders. Engineers secure APIs, implements system user authentication (JWT), configures third-party pipelines (Stripe, shipping calculators), and establishes cloud hosting architecture.
- **Key Deliverable:** REST or GraphQL APIs, encrypted database infrastructure, and functional CI/CD pipeline automation.

2.5 Quality Assurance (QA) Engineer

- **Responsibilities:** Writes test cases to catch visual and functional bugs across multiple web browsers. Executes end-to-end testing scripts on the payment checkout gateway, and validates non-functional security and platform traffic requirements.
- **Key Deliverable:** Automated test suites, bug tracking logs, and a final compliance verification sign-off sheet.

3. Resource Responsibility Assignment Matrix (RACI)

This matrix maps out how project members interact with major project deliverables:

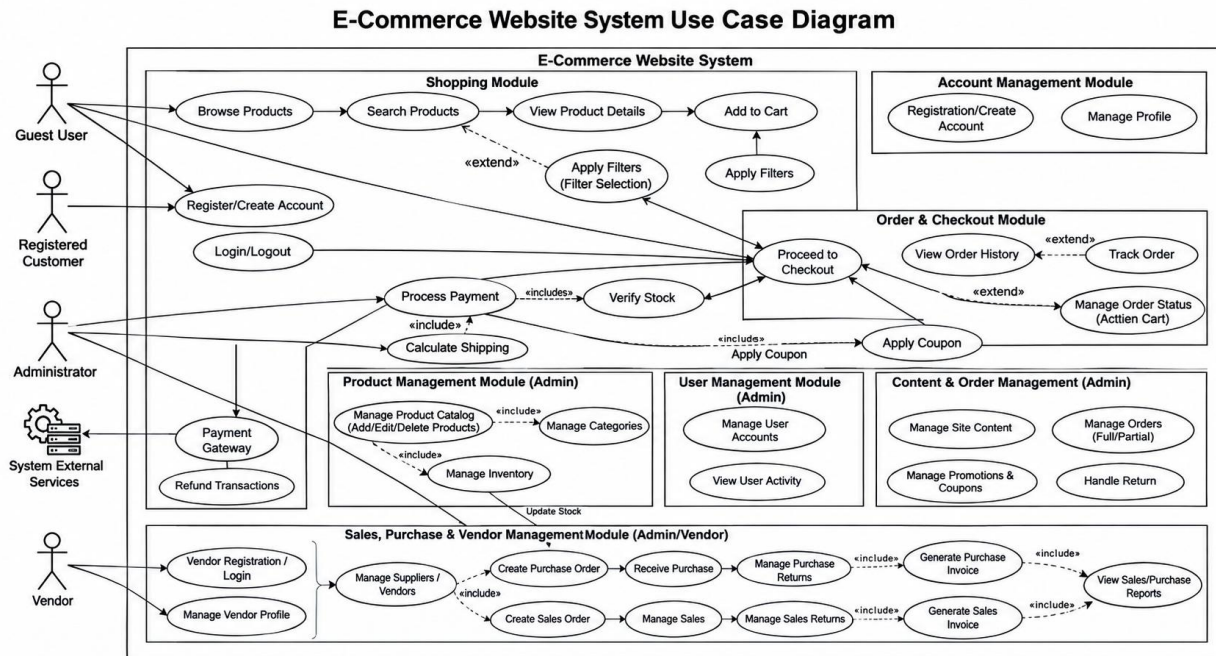
- **R (Responsible):** The person who physically executes the task.
- **A (Accountable):** The person with final decision-making power and ownership over success.
- **C (Consulted):** Subject matter experts whose input is gathered prior to work.
- **I (Informed):** Stakeholders kept up to date on progress but not actively involved.

Deliverable / Task	Project Manager	UI/UX Designer	Frontend Eng.	Backend Eng.	QA Engineer
SRS Document Sign-Off	A	C	C	C	C
UI/UX Wireframes	I	R / A	C	I	I
Database Schema Design	I	I	C	R / A	I
Payment Gateway Checkout	I	C	R	R / A	I
End-to-End System Testing	I	I	C	C	R / A

Production Launch	A	I	R	R	R
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MUHAMMAD WASEEM AKHTAR, MUHAMMAD ALI DANISH & JASTY MARRY (BSCS FINAL YEAR PROJECT REPORT)

5.0 USE CASE DIAGRAM:



This **UML Use Case Diagram** details the functional requirements of an E-Commerce Website System. It illustrates how different types of users (actors) interact with various sections (modules) of the online store.

Here is a comprehensive breakdown of the components and workflows shown in the diagram:

1. Actors (System Users)

The actors on the left side represent the external entities that interact with the system:

- **Guest User:** An unauthenticated visitor who can explore products and initiate a purchase.
- **Registered Customer:** A logged-in shopper with elevated privileges (like viewing order history).
- **Administrator:** An internal manager responsible for handling backend operations, settings, and users.
- **System External Services:** Third-party integrations, specifically a **Payment Gateway** to handle transactions and refunds.
- **Vendor:** A third-party seller who manages their own profile, inventory, and b2b sales/purchases.

2. Core Functional Modules

The large rectangle defines the boundaries of the **E-Commerce Website System**, which is broken down into specialized modules:

Shopping Module

This covers the basic catalog exploration functions available to buyers:

- Users can **Browse Products**, **Search Products**, and **View Product Details**.
- Searching has an <<extend>> relationship with **Apply Filters (Filter Selection)**, meaning filtering is an optional feature during a search.
- From product details, users can **Add to Cart**, which links to a secondary **Apply Filters** function (likely for choosing product variations like size or color).

Account Management Module

- Handles basic customer profile operations: **Registration/Create Account** and **Manage Profile**.
- Note that a **Guest User** is pointing to **Register/Create Account**, allowing them to convert into a **Registered Customer**.

Order & Checkout Module

This is the transactional hub where the purchase is completed:

- Both Guest and Registered users can **Proceed to Checkout**.
- **Registered Customers** can uniquely **View Order History**, which can optionally be extended (<<extend>>) to **Track Order**.
- **Proceed to Checkout** triggers a series of essential, mandatory actions indicated by <<includes>> relationships:
 - **Process Payment** (which communicates externally with the **Payment Gateway**).
 - **Verify Stock**.
 - **Calculate Shipping**.
 - **Apply Coupon** (optional extension via <<extend>> from the Checkout flow).

Administrative Modules (Admin & Vendor)

These modules handle the backend management of the platform:

- **Product Management Module (Admin):** Allows admins to **Manage Product Catalog** (Add/Edit/Delete). This mandatory process includes (<<include>>) **Manage Categories** and **Manage Inventory**.
- **User Management Module (Admin):** Allows admins to **Manage User Accounts** and **View User Activity**.
- **Content & Order Management (Admin):** Used to **Manage Site Content**, **Manage Orders**, **Manage Promotions & Coupons**, and **Handle Return** requests.
- **Sales, Purchase & Vendor Management Module:** A shared backend portal for **Vendors** and **Admins** to handle supply chains. It features login registration, vendor profile management, and a sequential workflow for sales/purchase pipelines (e.g., creating purchase orders, generating invoices, and viewing sales reports).

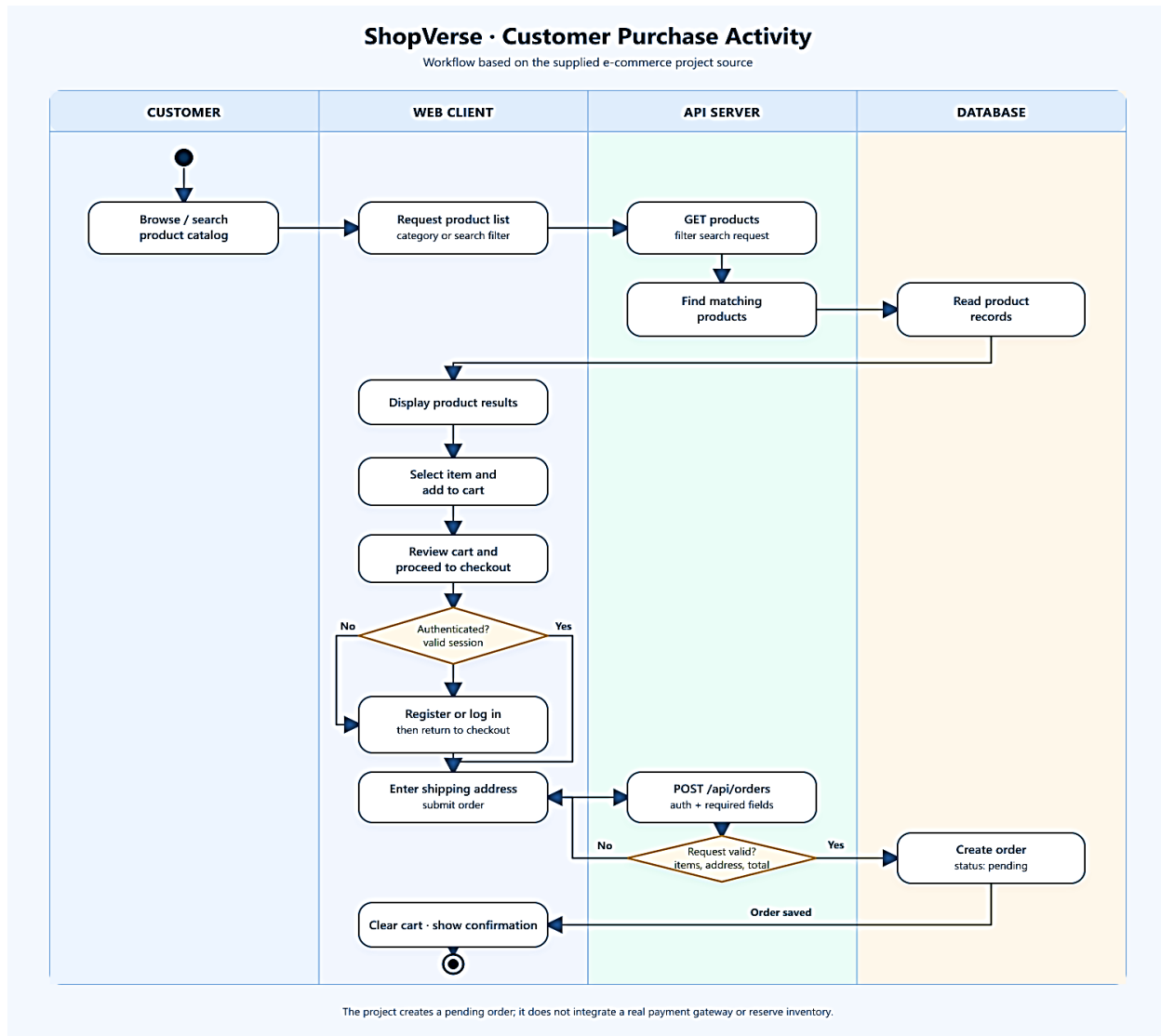
4. Key Relationships Explained

- **Solid Lines (Associations):** Connect actors directly to use cases, indicating who has permission to perform that action (e.g., Administrator connects to Process Payment).

- **<<include>> Arrow:** Means the base use case **requires** the functional steps of the pointed usecase to complete successfully (e.g., You cannot finish "Process Payment" without also executing "Calculate Shipping").
- **<<extend>> Arrow:** Means the pointed use case is an **optional** feature that can happen under certain conditions (e.g., A user searching for a product *might* choose to apply filters, but it isn't mandatory).

Would you like me to focus on a **specific workflow** within this diagram (such as the detailed checkout sequence), or do you need help **translating this into software code requirements**?
AI can make mistakes, so double-check responses

6.0 ACTIVITY DIAGRAM:

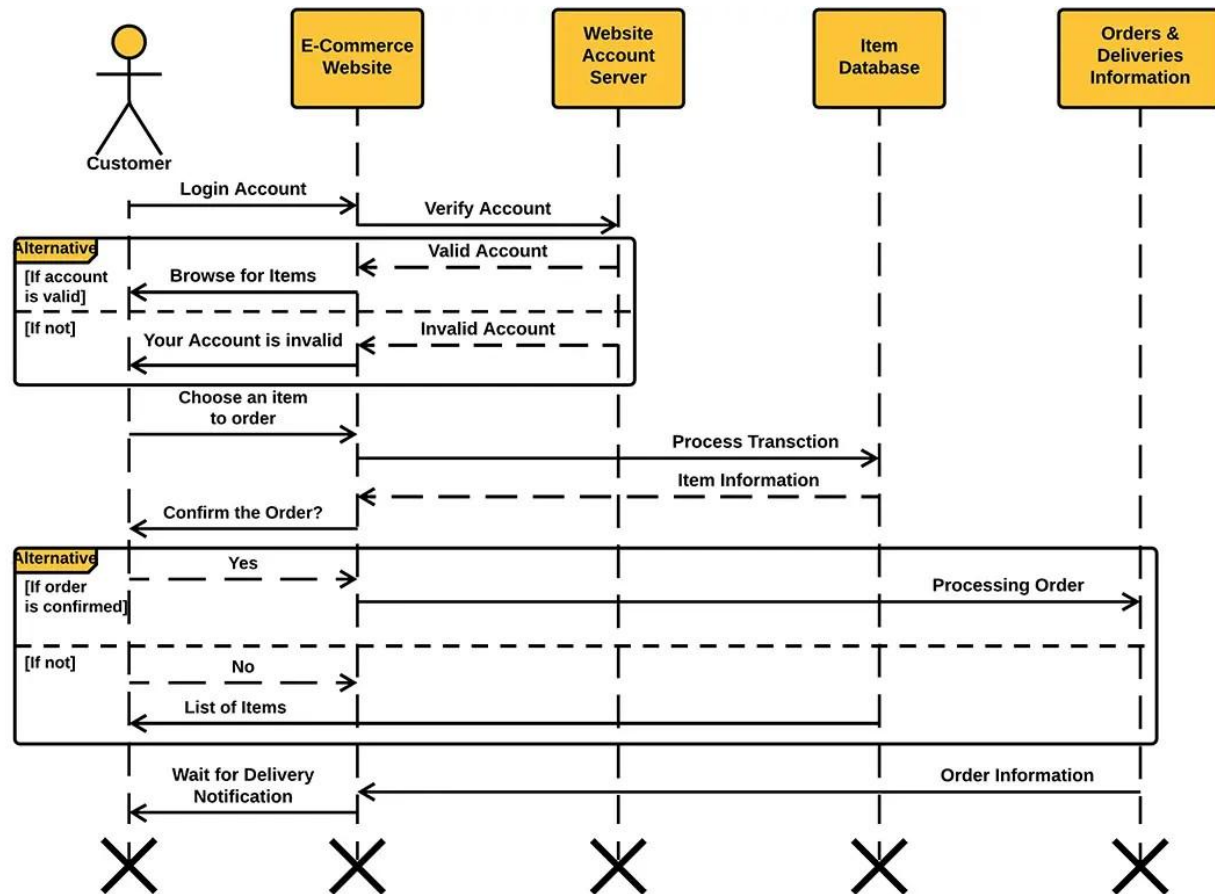


The customer shops and checks out. If needed, they log in or register. The server validates the order; if valid, the database saves it as pending, and the site confirms the order and clears the cart.

Symbol guide:

- **Start** ● Start the activity.
- **Rounded box** Perform an action, such as browsing products or submitting an order.
- **Arrow** → Shows the direction of the workflow.
- **Diamond** ◇ Check a condition and choose a path. Here: Is the customer logged in? Is the order request valid?
- **Yes / No** Labels the outcome of a decision.
- **Swimlanes** Show who or what performs each step: Customer, Web Client, API Server, or Database.
- **End** ◎ End the activity.

7.0 SEQUENCE DIAGRAM:



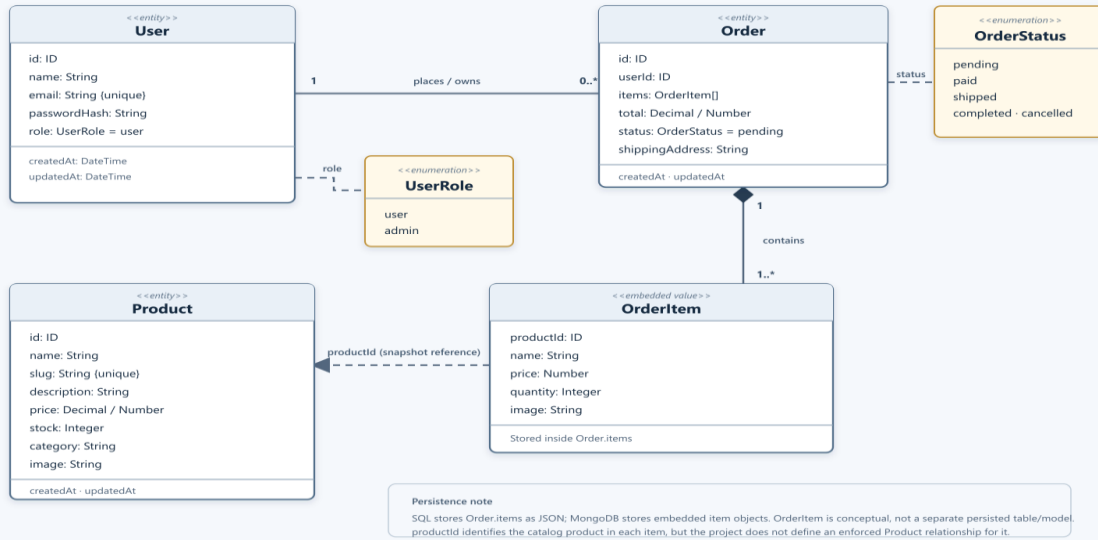
Key points:

- The **customer logs in**; the account server verifies the account.
- If the account is valid, the customer browses items. If it is invalid, the customer is notified.
- The customer selects an item, and the website retrieves its details from the item database.
- The customer chooses whether to confirm the order.
- If confirmed, the order is processed and delivery information is sent.
- If not confirmed, the website returns to the item list.
- The customer waits for a delivery notification.

8.0 CLASS DIAGRAM:

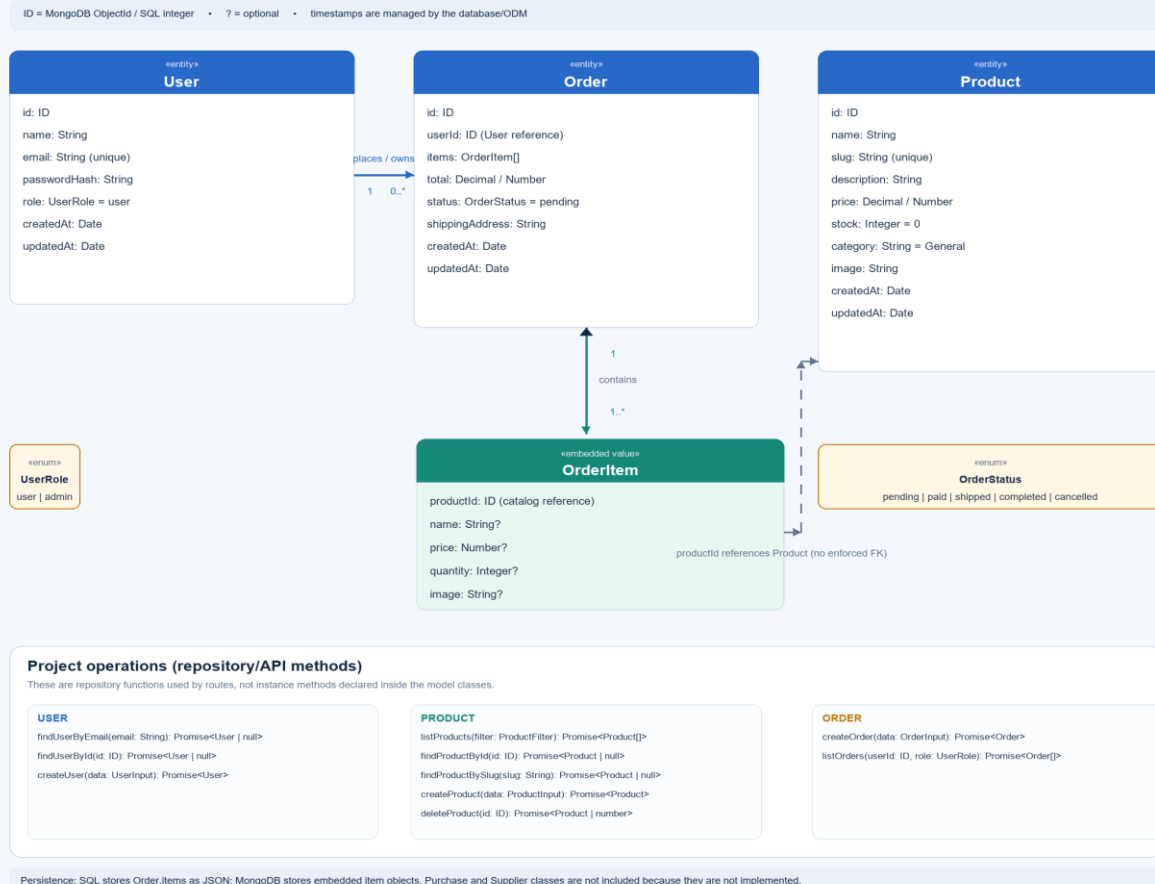
ShopVerse · Class Diagram

Core data classes from the supplied e-commerce project



ShopVerse — Class Diagram

Typed properties and project operations for the core e-commerce data classes



Relationships:

1 User → many Orders. 1 Order → 1 or more OrderItems. Each order item includes a product ID, but it is not enforced as a direct database relationship.

Class diagram in brief

- **User** stores account details and a role: user or admin.
- **Product** stores catalog details, price, stock, category, and image.
- **Order** records the customer, items, total, status, and shipping address.
- **OrderItem** stores each ordered product's details and quantity inside an order.
- **OrderStatus** can be pending, paid, shipped, completed, or cancelled.

ShopVerse Class Diagram — Presentation Brief

Overview

This class diagram shows the main data structures used by ShopVerse, an e-commerce website: **User**, **Product**, **Order**, and **OrderItem**. It also shows the UserRole and OrderStatus enumerations and the repository operations used to work with these records.

A useful way to explain the diagram is to follow the customer journey: a user views products, places an order, and the system stores the order with its items and status.

Main classes

1. User

The User class stores customer and administrator account information.

- id identifies the user.
- name and email hold account details. Email must be unique.
- passwordHash stores the protected password value, not the plain-text password.
- role indicates whether the account is a regular user or an admin.
- createdAt and updatedAt record when the account was created and last updated.

Relationship: One user can place zero or many orders. Each order belongs to one user.

2. Product

The Product class represents an item available in the store.

- id identifies the product.
- name, slug, and description provide its identifying and display information. The slug is unique.
- price is the selling price.
- stock records the available quantity.
- category groups similar products.
- image stores the product image reference.
- createdAt and updatedAt track record changes.

The project supports listing and searching products, finding them by ID or slug, adding products, and deleting products.

3. Order

The Order class records a customer's checkout.

- id identifies the order.
- userId links the order to its customer.
- items contains the products and quantities in the order.
- total stores the order amount.
- status tracks the order's progress.
- shippingAddress stores where the order should be delivered.
- createdAt and updatedAt record order timestamps.

When an order is created, the project sets its initial status to pending. A user can view their own orders; an administrator can view all orders.

4. OrderItem

OrderItem describes one product entry inside an order. It includes a product ID, product name, price, quantity, and image reference.

The project stores order items **inside the order**: as JSON in SQL or as embedded objects in MongoDB. OrderItem is therefore a conceptual embedded value in this diagram, not a separate persisted table or collection. Its productId identifies the catalog product, but the project does not enforce a database foreign-key relationship for it.

Relationships and enumerations

- **User → Order**: One user can have multiple orders; each order is linked to one user.
- **Order → OrderItem**: An order contains one or more item entries.
- **OrderItem → Product**: An item references a product in the catalog.
- **UserRole**: The allowed roles are user and admin.
- **OrderStatus**: The allowed statuses are pending, paid, shipped, completed, and cancelled.

Operations shown in the diagram

The operation signatures are grouped by User, Product, and Order. They are **repository/API functions**, rather than methods defined directly on the model classes.

- **User operations** find a user by email or ID and create a user.
- **Product operations** list and filter products, find a product by ID or slug, create a product, and delete a product.
- **Order operations** create an order and list orders according to the requesting user's role.

The Promise<...> notation means the operation returns its result asynchronously. For example, Promise<Product[]> means it returns a list of products.

Types and notation

- String represents text; Integer represents a whole number.
- Decimal / Number represents a monetary or numeric value. The exact database type varies between SQL and MongoDB.
- Date represents a timestamp.

- ID represents the record identifier: an integer in SQL or an ObjectId in MongoDB.
- ? marks an optional property.
- [] means a list, such as OrderItem[] for the items within an order.

Short presentation script

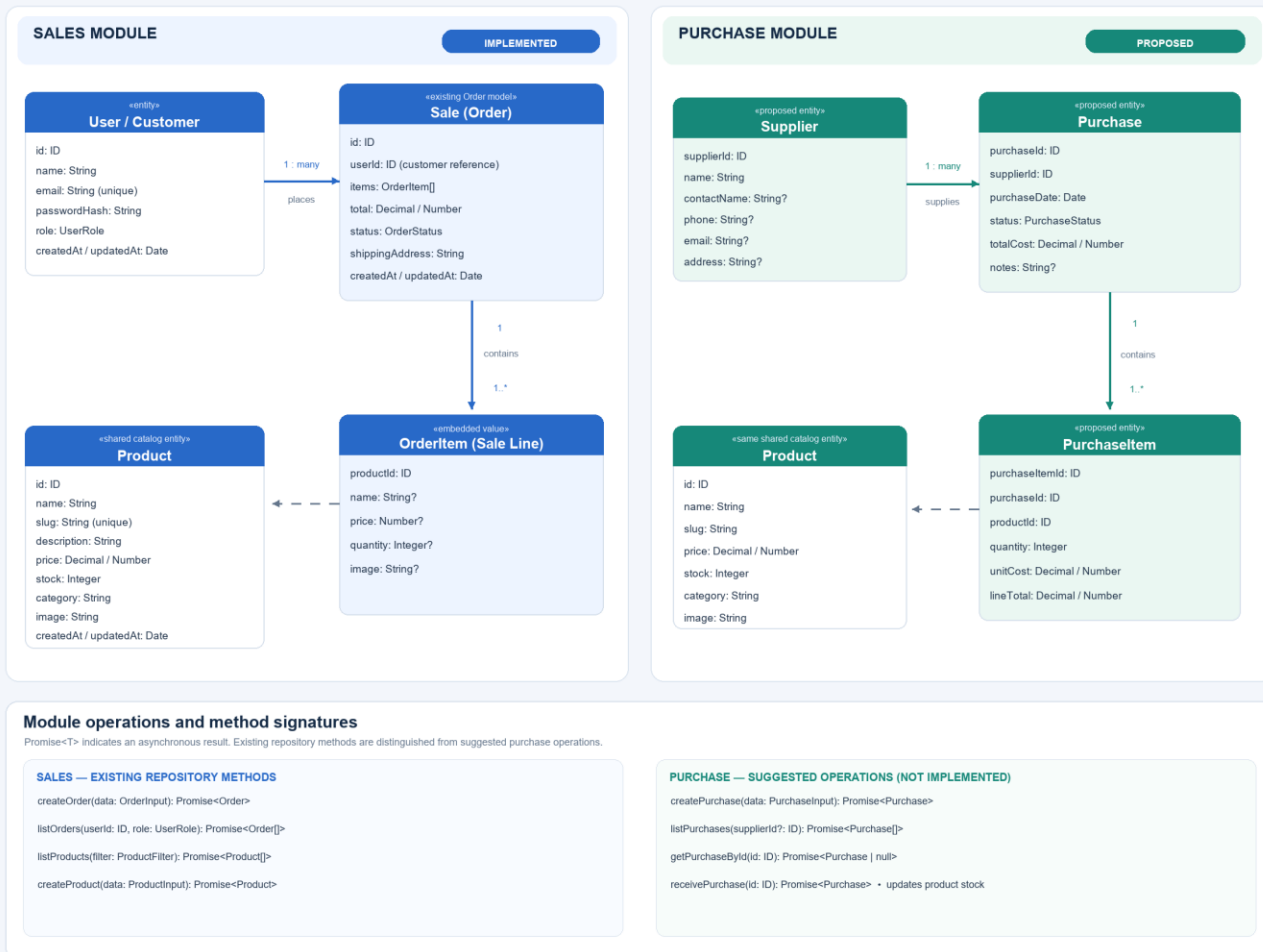
“This diagram explains how ShopVerse organizes its core e-commerce data. A user can place multiple orders, and each order stores its total, shipping address, status, and item details. Each item refers to a product in the catalog. User roles distinguish customers from administrators, while order statuses track progress from pending through completion or cancellation. The methods shown are repository operations that the application uses to find, create, list, and delete records. Finally, order items are embedded in an order in both supported database approaches, rather than saved as independent records.”

Sales & Purchase Class Diagram

ShopVerse — Sales & Purchase Class Diagram

Typed entities, relationships, and operations • Based on the supplied e-commerce project

ID = MongoDB ObjectId / SQL integer • ? = optional • Purchase is proposed • dashed link = logical reference



Important: Current sales are represented by the project's Order model. Supplier, Purchase, PurchaseItem, and purchase methods are proposed additions.

ShopVerse Sales and Purchase Modules — Presentation Brief

Overview

This diagram presents two parts of the ShopVerse e-commerce system: **Sales**, which is implemented through the existing order model, and **Purchase**, a proposed module for recording goods bought from suppliers.

The Product class appears in both panels for readability. It represents the same shared product catalog.

Sales module — implemented

1. A customer, represented by a User account, places an order.
2. The order stores its customer ID, items, total, shipping address, status, and timestamps.
3. Each OrderItem records a product ID and may include a name, price, quantity, and image.
4. Order items are embedded inside the order: SQL stores them as JSON, while MongoDB stores embedded objects.
5. The item's product ID refers to a catalog Product. The dashed line indicates a logical reference; the project does not enforce a database foreign key for it.
6. The order starts with pending status and can be marked paid, shipped, completed, or cancelled.

Existing operations include creating an order, listing orders, listing products, and creating products. Order listing returns a customer's own orders, while administrators can see all orders.

Purchase module — proposed

1. A Supplier provides products to the store.
2. A Purchase record captures the supplier, purchase date, status, total cost, and optional notes.
3. Each purchase contains one or more PurchaseItem records.
4. Each PurchaseItem links to a Product and stores the quantity, unit cost, and line total.
5. A proposed receivePurchase operation could update product stock when the purchased goods arrive.

The supplier, purchase, purchase-item records, and purchase operations are not part of the supplied project yet. They are shown as a suggested extension.

Key relationships

- One user can have multiple orders.
- One order contains one or more order items.
- One supplier can have multiple purchases.
- One purchase contains one or more purchase items.
- Both sales and purchasing refer to the shared Product catalog.

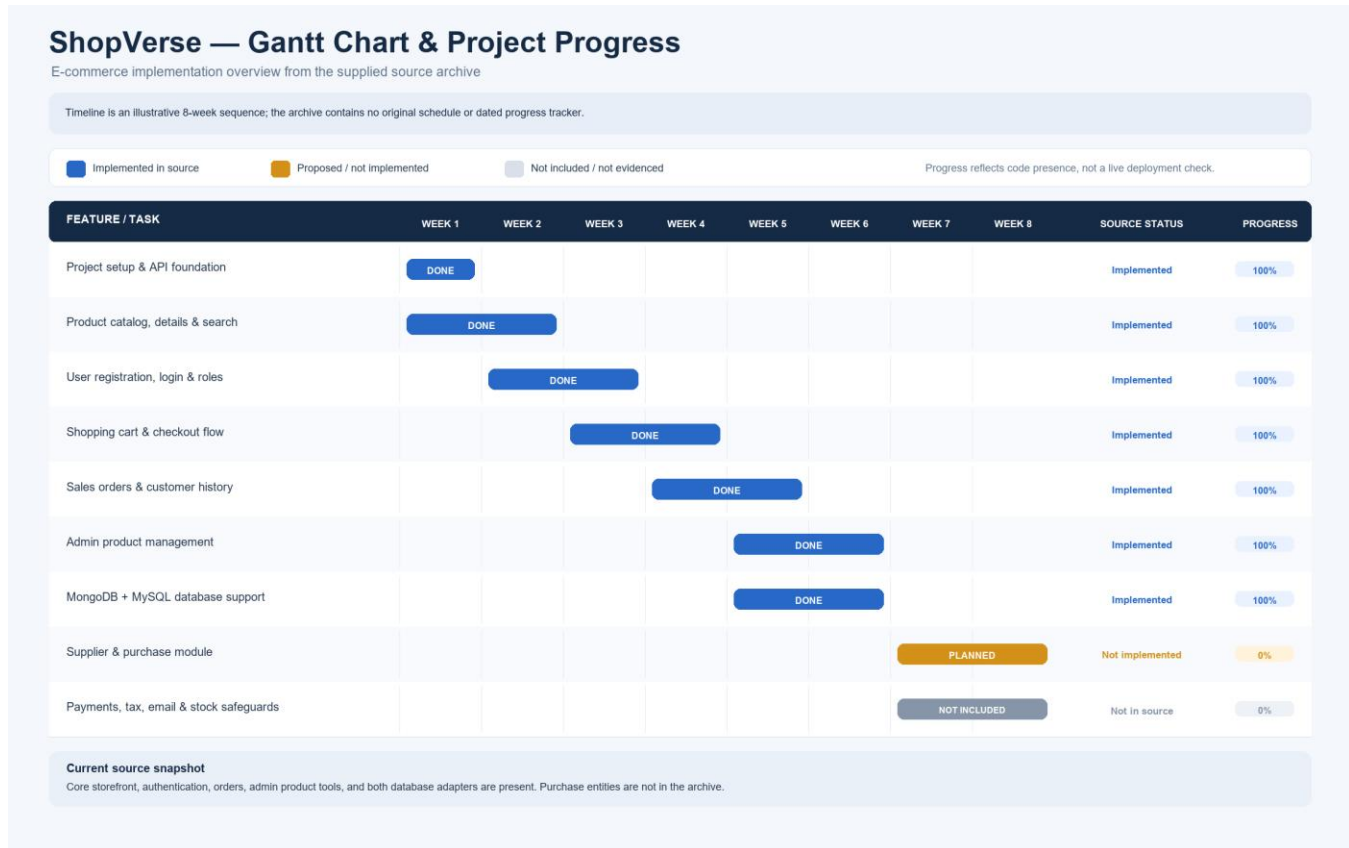
Types and notation

ID represents a MongoDB ObjectId or SQL integer. String, Integer, Date, and Decimal / Number indicate the expected value types. A ? marks an optional field. Promise<T> means an operation returns its result asynchronously. Dashed lines show logical references.

Short presentation script

“This diagram separates ShopVerse’s sales process from a proposed purchasing process. The sales side is already implemented: a user places an order, the order contains embedded item details, and each item references a product in the catalog. The order also tracks the total, delivery address, and status. The purchase side extends the system to record suppliers and stock purchases. Each purchase can contain multiple purchase items, including quantity and unit cost. Once a purchase is received, the proposed process can update product stock. The dashed links indicate product references, and the purchase module is clearly marked as a future addition.”

9.0 GANTT CHART PROGRESS:



What the chart shows

This Gantt-style chart summarizes the main ShopVerse development areas and their status in the supplied source archive. Blue bars indicate features found in the source. The purchase module is marked as proposed because it is not implemented. Payment, tax, email, and stock safeguards are marked as not included.

The week labels show an **illustrative development sequence**, not actual project dates. The archive contains no dated schedule or progress tracker. “100%” means the related code is present in the archive; it does not confirm that the feature has been tested or deployed.

Progress, step by step

1. Project setup and API foundation — implemented

The source includes the server application and API structure needed to run the e-commerce site.

2. Product catalog, details, and search — implemented

Users can browse product information, search, and filter products.

3. **User registration, login, and roles — implemented**

The project supports user accounts and distinguishes regular users from administrators.

4. **Shopping cart and checkout — implemented**

The client includes cart and checkout flows for preparing an order.

5. **Sales orders and customer history — implemented**

Checkout creates orders. The project stores order items, totals, shipping addresses, and statuses, and provides order history.

6. **Admin product management — implemented**

Administrators can add and delete products.

7. **MongoDB and MySQL support — implemented**

The source includes models and configuration for either database option.

8. **Supplier and purchase module — proposed**

The archive does not contain supplier or purchase records. This is shown as future work for recording suppliers, purchases, purchase items, quantities, and costs.

9. **Payments, tax, email, and stock safeguards — not included**

The project README identifies these as production features that still need to be added.

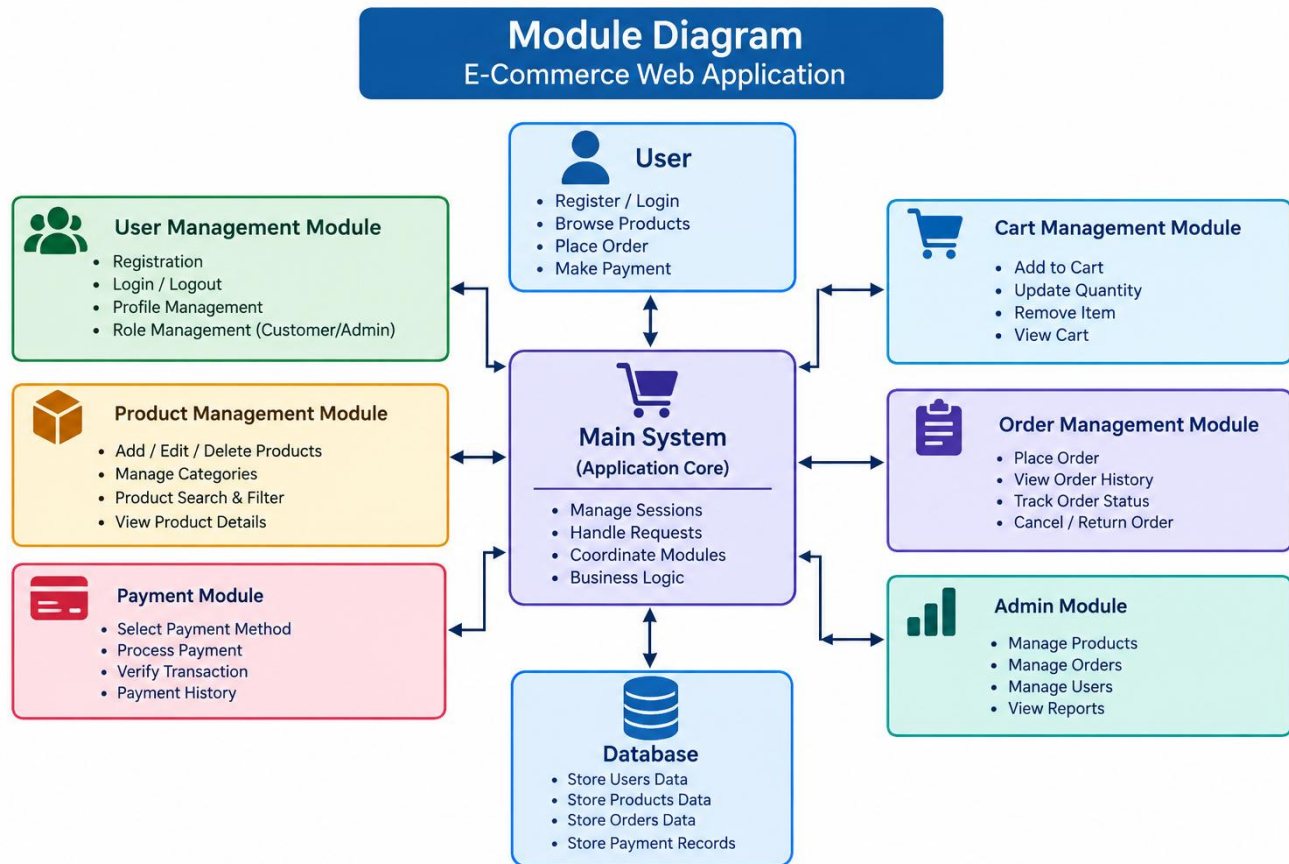
Key points to present

- The core storefront, user accounts, product catalog, checkout, sales orders, admin product tools, and database options are represented in the source.
- The purchase module is an extension idea, not a completed feature.
- The chart's week-by-week bars illustrate a possible sequence; they are not a record of when the work was actually completed.
- The percentages reflect whether related code appears in the archive. Testing, deployment, and live operation were not verified.

Short presentation script

"This Gantt chart gives an overview of ShopVerse's development status based on the source files. The implemented core includes the product catalog, account management, cart and checkout, sales orders, admin product management, and support for MongoDB or MySQL. The supplier and purchase module is proposed future work. Payment, tax, email, and some stock safeguards are also not included in the current project. Since the archive doesn't provide dates or a progress tracker, the week sequence is illustrative, and the percentages indicate code presence rather than tested or deployed completion."

10.0 MODULE DIAGRAM:



The **Module Diagram** represents the main functional components of the **E-Commerce Web Application** and shows how they interact with the central application system and database.

Key Points

- **User Module:** Handles user registration, login/logout, profile management, and customer/admin roles.
- **Product Management Module:** Manages products, categories, product details, searching, editing, and deletion.
- **Cart Management Module:** Allows users to add products, update quantities, remove items, and view their cart.
- **Order Management Module:** Handles order placement, order history, status tracking, cancellation, and returns.
- **Payment Module:** Manages payment methods, payment processing, transaction verification, and payment history.
- **Admin Module:** Allows administrators to manage products, users, orders, and reports.

- **Main System:** Acts as the core of the application, coordinating all modules, handling requests, sessions, and business logic.
- **Database:** Stores important information such as users, products, orders, and payment records.
- **Module Interaction:** The arrows show communication and data flow between the user, application modules, main system, and database.

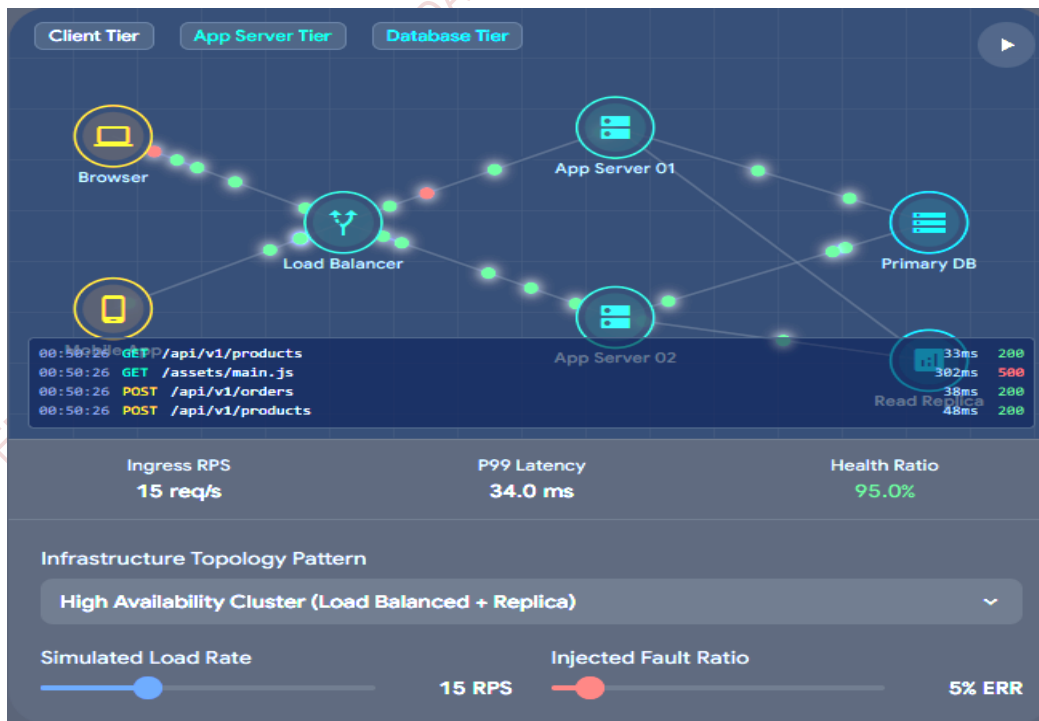
Overall Purpose

The diagram provides a **clear high-level view of the system architecture**, making it easier to understand how different e-commerce functions are divided into separate modules and how they work together.

11.0 ARTITECTURE DIAGRAM:



Monolith Setup (Single Server / No Replica)



High Availability Cluster (Load Balanced + Replica)



Cached Accelerated Cluster (Redis + Read Replicas)

Based on the provided project file log, here is a comprehensive breakdown of the application's software architecture.

The system implements a classic **decoupled Client-Server (Three-Tier) Architecture** with an independent database management tier.

1. Client Tier (Frontend)

The frontend utilizes a single-page or lightly routing multi-page interface powered by raw JavaScript, semantic HTML5, and CSS.

text

ecommerce-website/client/

```

|— css/
|   |— style.css      # Central global styling
|— js/
|   |— api.js         # Central API client (Fetch/Axios wrappers for backend communication)

```

```

|   |— main.js           # Global scripts / initialization
|   |— auth.js          # State logic for Login/Registration
|   |— products.js      # Logic for viewing lists and catalogs
|   |— product.js       # Detailed single product views
|   |— cart.js          # Local storage or session management of current items
|   |— checkout.js      # Order placements and multi-step forms
|   |— account.js       # User profiles and order histories
|   |— admin.js         # Administrative panel scripts
|— index.html           # Main landing page
|— login.html / register.html
|— products.html / product.html
|— cart.html / checkout.html
|— account.html         # Customer dashboard
|— admin.html           # Admin portal interface

```

Use code with caution.

2. Server Tier (Backend App)

The server relies on a monolithic **Node.js/Express** app that separates concerns using modular routing and structural layers.

ecommerce-website/server/

```

|— src/
|   |— config/
|   |   |— db.js        # Initialized configuration setups for DB connections
|   |— middleware/
|   |   |— auth.js      # Request authorization (e.g., JWT verifying token filters)
|   |— routes/          # Explicit API endpoint groups
|   |   |— auth.js      # Auth-related controllers (/api/auth)
|   |   |— products.js  # Product and catalog manipulation operations (/api/products)
|   |   |— orders.js    # Finalization of transactions and checkouts (/api/orders)
|— .env.example         # Template for infrastructure-specific variables
|— app.js               # Application bootstrap / Middleware registry / Port listener
|— seed.js              # Independent execution script for local data populations
|— package.json         # Manifest declaring server-side dependencies

```

Use code with caution.

3. Data Tier (Persistence Layer)

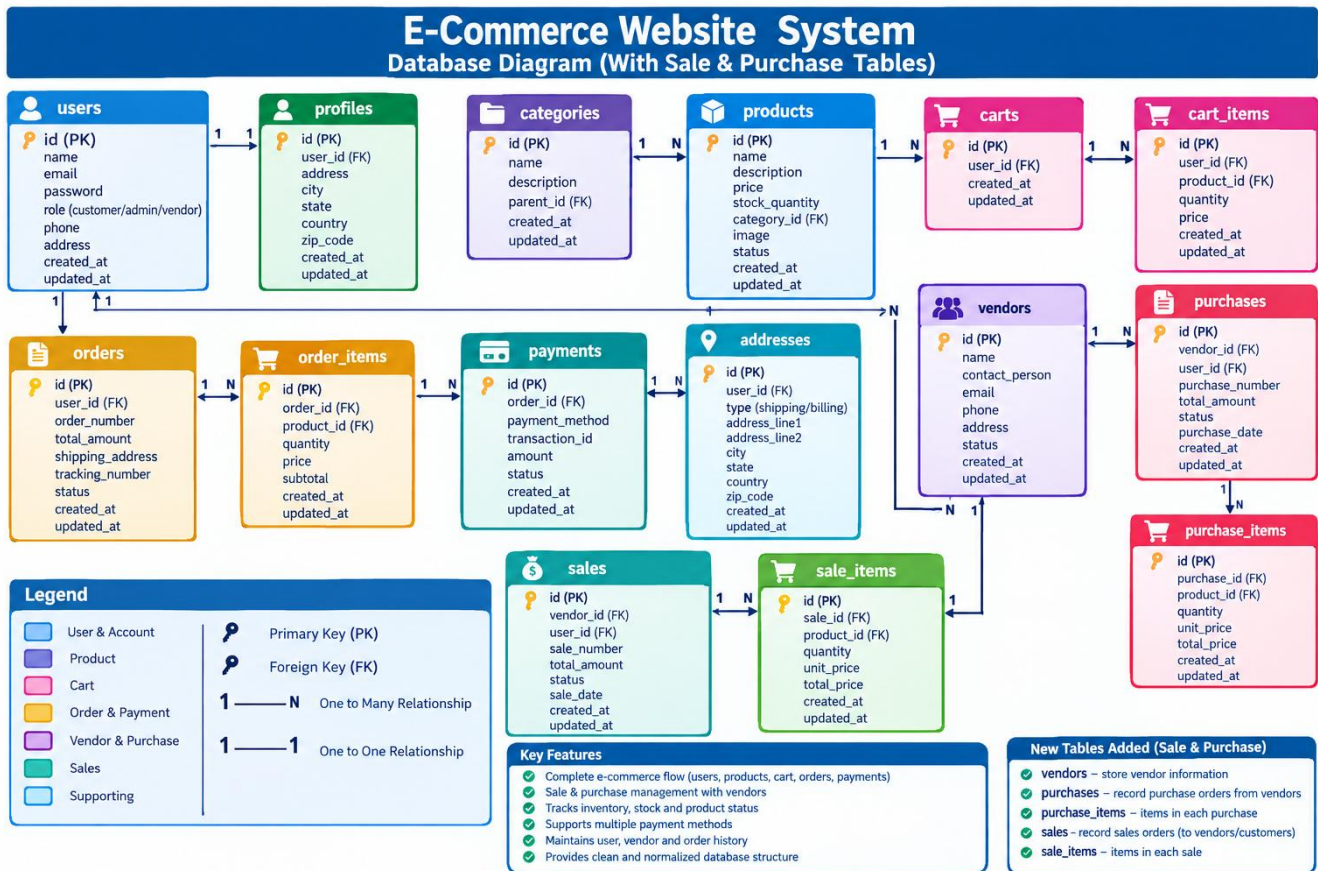
The persistence codebase hints at a flexible, hybrid, or dual database strategy supporting both **SQL** and **NoSQL (MongoDB)** models.

```
ecommerce-website/
├── database/
│   ├── sql/
│   │   ├── schema.sql    # Structured relational table statements (DDL)
│   │   └── seed.sql      # Direct SQL records for database baseline population
│   └── server/src/db/
│       ├── models-sql.js  # Object-Relational Mapping (ORM) setup (e.g., Sequelize models)
│       └── models-mongo.js # Object-Document Mapping (ODM) configurations (e.g.,
Mongoose models)
│           └── repository.js # Repository Pattern implementation (abstracts query mechanisms
away from routes)
Use code with caution.
```

Architectural Pattern Analysis

- **Separation of Concerns:** The architecture segregates presentation layout (client/*.html), interactive behavioral scripts (client/js/*), business endpoints (server/src/routes/*), security controls (server/src/middleware/*), and structural query code (server/src/db/*).
- **Repository Pattern:** The introduction of repository.js creates an abstraction layer. This completely insulates the API route endpoints from whether the application is actively saving records to the SQL database (models-sql.js) or the Mongo instance (models-mongo.js).

12.0 ADD SALE PURCHASE TABLES:



This image illustrates an Entity-Relationship (ER) database schema blueprint tailored for an **E-Commerce Website System**. It details how data flows and connects between different aspects of an online retail platform, including user accounts, inventory tracking, shopping carts, and financial management.

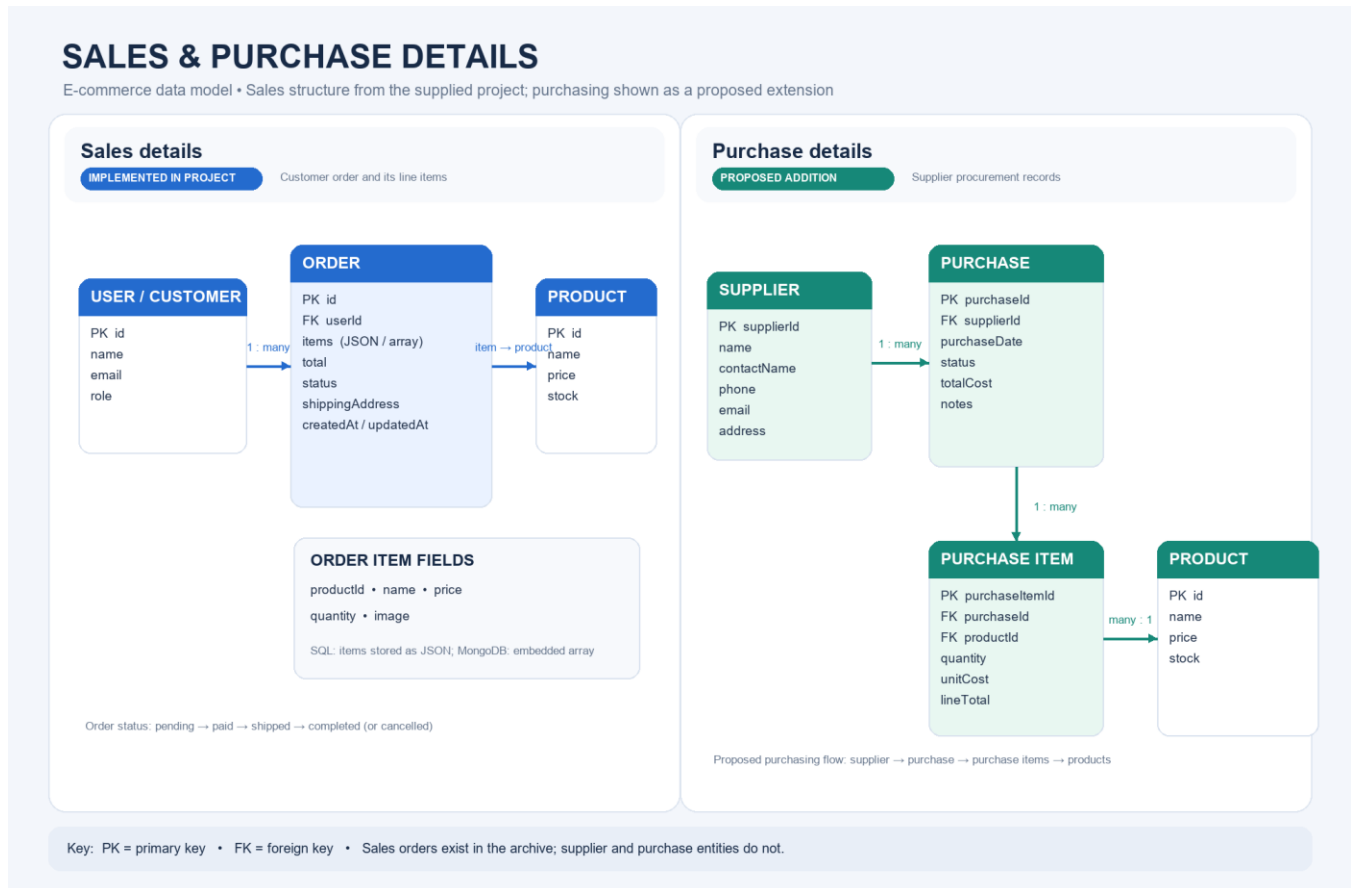
Key Functional Sections

- User & Account Management:** The users, profiles, and addresses tables store essential credentials, user roles (customer, admin, vendor), contact numbers, and delivery locations.
- Product Inventory:** The categories and products tables organize items, track available stock quantities, map out hierarchy levels (parent_id), and display item statuses.
- Shopping Cart & Checkout:** The carts and cart_items tables act as temporary holding areas for customer selections before purchase processing.
- Orders & Payments:** The orders, order_items, and payments tables log finalized customer receipts, item breakdowns, pricing calculations, and transaction references.
- Vendor Operations (New Additions):** The vendors, purchases, purchase_items, sales, and sale_items tables trace standard business-to-business inventory intake alongside merchant sales metrics.

System Relations & Design

- **Primary (PK) & Foreign Keys (FK):** Every table employs an id field as a unique primary key identifier, linking seamlessly to other tables via explicit foreign keys to keep data unified.
- **Cardinality:** The schema utilizes explicit **One-to-Many (1 $\rightarrow$ N)** and **One-to-One (1 $\rightarrow$ 1)** structural connections to keep records relational and clean.

13.0 SALES AND PURCHASE DETAILS:



Brief explanation: The sales section shows how a customer's order is recorded. One customer can have many orders, and each order stores its items, total, shipping address, and status. The order items refer to products and include the product name, price, and quantity.

The purchase section shows a suggested way to track stock bought from suppliers. One supplier can have many purchases, and each purchase can contain multiple purchase items linked to products. Purchase records are a proposed addition; they are not currently implemented in the project.

Sales Details

The **Sales Details** section represents customer orders. Each customer can place multiple orders, and each order stores the customer ID, ordered items, total amount, shipping address, current status, and timestamps. The order items include product information such as product ID, name, price, and quantity. In the project, these items are stored inside the order as JSON in SQL or as an embedded array in MongoDB.

Purchase Details

The **Purchase Details** section is a proposed way to record products bought from suppliers. Each supplier can have multiple purchases, and each purchase can contain several purchase items. Each

item links a product to the purchase and records its quantity and cost, helping calculate the total purchase cost.

Key Points Sale & Purchase Details

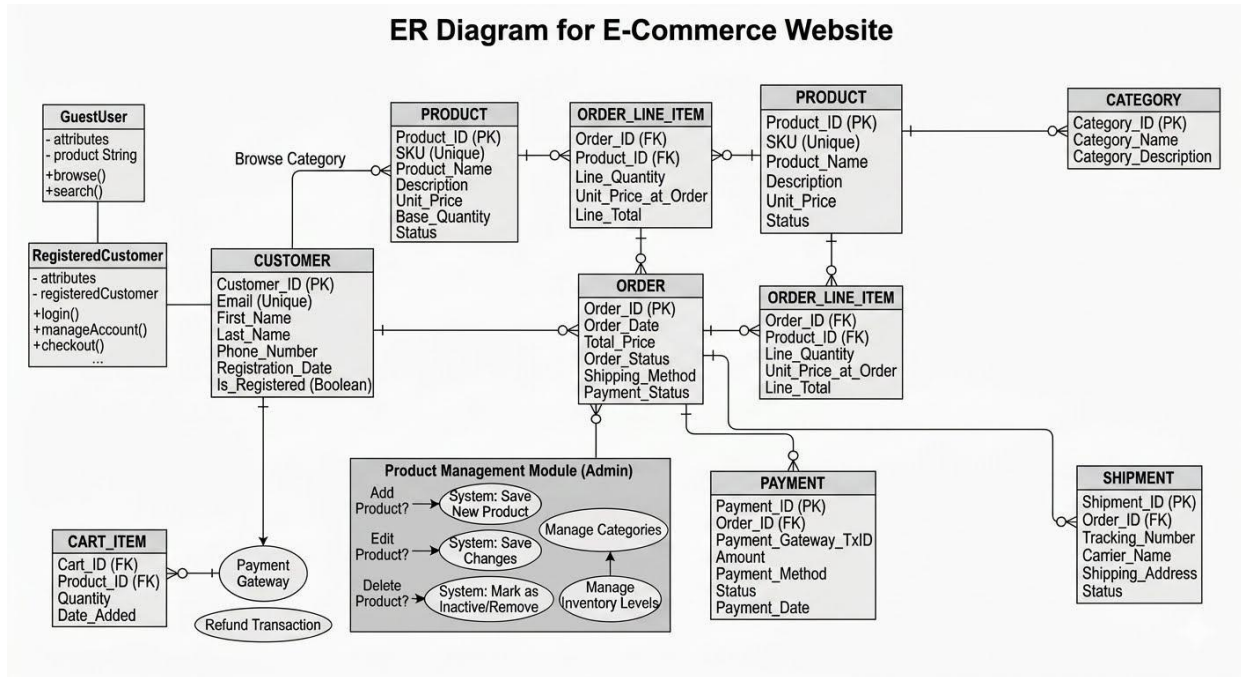
Sales process — implemented in the project

1. A customer places an order.
2. The order is linked to the customer through `userId`.
3. The order stores the items, total, shipping address, status, and timestamps.
4. Each item records product details such as its ID, name, price, and quantity. Items are stored inside the order as JSON in SQL or an embedded array in MongoDB.
5. The order status can move from **pending** to **paid**, **shipped**, and **completed**, or become **cancelled**.

Purchase process — proposed addition

1. A supplier provides products to the store.
2. A purchase record is created and linked to the supplier.
3. Each purchase contains one or more purchase items.
4. Each purchase item links to a product and records the quantity and unit cost.
5. The purchase total is calculated from its items.

14.0 ER DIAGRAM:



ER diagram:

key points

- **Customer** information is stored in the CUSTOMER entity. Guest and registered users represent different ways customers use the site.
- **Products** are organized into categories. Each product has details such as SKU, name, price, and status.
- **Orders** belong to customers. ORDER_LINE_ITEM connects orders to products and records quantity and price.
- **Cart items** hold products and quantities before checkout.
- **Payments** and **shipments** are associated with orders and store payment or delivery details.
- Admin features manage products, categories, and inventory.

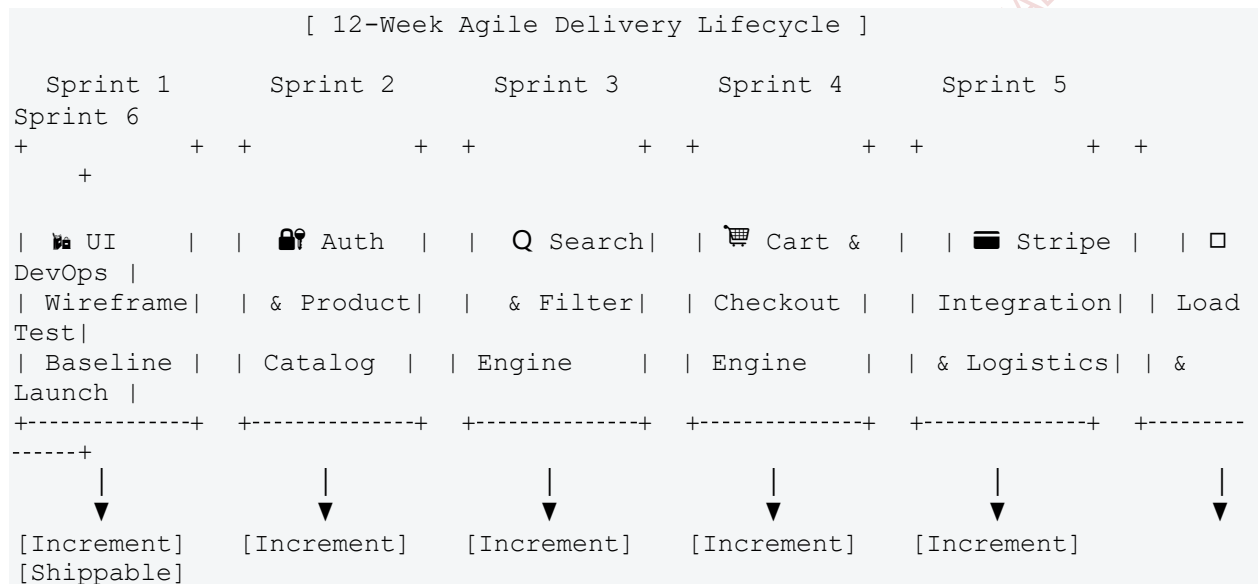
How to read it: boxes are entities; listed fields are their data; (PK) marks a primary key, (FK) a foreign key; lines show relationships.

15.0 PROCESS MODEL, AGILE:

This **Agile Process Model** details how the E-Commerce Website project will be executed dynamically across interactive iterations. It outlines the specific artifacts, sprint rhythms, and user stories required to build and deploy the platform incrementally.

1. The E-Commerce Sprint Strategy

The development workload is broken down into **six consecutive 2-week sprints** using an incremental process model. Rather than launching all components at once, each sprint delivers a testable, production-ready slice of the platform.



2. Core Scrum Ceremonies for the Project

To maintain development velocity, the cross-functional team adheres to a strict sprint battle rhythm:

- **Sprint Planning (First Monday of Sprint, 2 hours):** The Product Owner defines high-priority items from the product backlog. The engineers break down these items into tasks and commit to a specific sprint goal (e.g., *"Enable secure credit card processing via Stripe"*).
- **Daily Scrum (Mon–Fri, 15 minutes):** A timeboxed sync where developers answer three core questions:
 1. What did I complete yesterday to help meet the sprint goal?
 2. What will I work on today?
 3. Are there any technical blockers in my way?
- **Sprint Review (Last Friday of Sprint, 1 hour):** The team demonstrates a live, working increment of the e-commerce site to stakeholders (e.g., demonstrating a functional user registration or cart page).

- **Sprint Retrospective (Last Friday of Sprint, 1 hour):** The Scrum Master facilitates a review session to determine what went well, what went wrong, and actions to improve workflow efficiency for the next sprint.

3. Product Backlog & Sample User Stories

The Product Owner manages the e-commerce features as discrete, customer-centric **User Stories**. Each story follows standard Agile syntax and must pass rigorous acceptance criteria to be marked as complete.

User Story 1: Persistent Shopping Cart (Sprint 4 Priority)

- **As a** registered customer,
- **I want to** add items to my shopping cart and close the browser,
- **So that** my items are still saved when I log back in later on another device.
- **Acceptance Criteria:**
 - Cart items must link to the user's database ID when authenticated.
 - Unauthenticated guest carts must persist via local storage tokens for 7 days.
 - The cart badge must update dynamically without forcing a full page refresh.

User Story 2: Secure Payment Processing (Sprint 5 Priority)

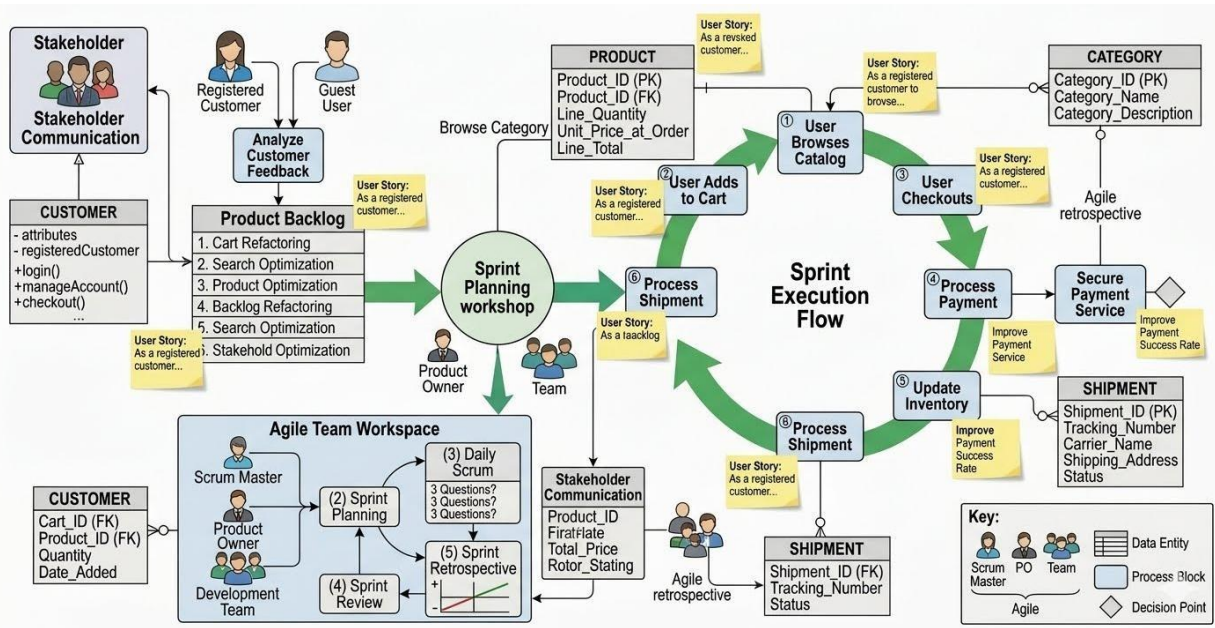
- **As a** buyer check out on the web storefront,
- **I want to** input my credit card credentials securely,
- **So that** I can confidently complete my financial transaction.
- **Acceptance Criteria:**
 - Credit card forms must use Stripe Elements to avoid handling raw financial data.
 - The system must handle payment failure errors gracefully and display helpful alerts.
 - On payment success, an entry must generate in the confirmation ORDER table and a email must trigger.

4. Definition of Done (DoD)

Before any e-commerce user story can transition from "In Progress" to "Done" at the end of a sprint, it must pass the following standardized quality gate:

- **Code Quality:** Peer reviewed by at least one other engineer with zero critical linter warnings.
- **Testing:** Unit test coverage meets or exceeds 80% for backend route modifications.
- **Security:** No hardcoded API keys; checkout processes conform to strict PCI compliance guidelines.
- **Accessibility:** Storefront interface components conform to WCAG 2.1 AA design contrast compliance.

4. Process Model and Agile for E-Commerce Platform Diagram:



16.0 GANTT CHART ROUTES:

This **Gantt Chart Schedule** details the production timeline, sequential task paths, and deployment milestones for the E-Commerce Website across a **12-week development lifecycle**.

1. 12-Week Visual Project Schedule

PHASE / TASK NAME	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	DEPENDENCY
PHASE 1: RESEARCH & UI													
1.1 SRS Document Lock	[==]												
1.2 Wireframes & Mockups		[=====]											
1.1													
PHASE 2: FOUNDATION													
2.1 DB Schema & Models			[==]										
2.2 Auth & JWT Logic				[==]									
2.1													
PHASE 3: CORE FE & BE													
3.1 Storefront Screens				[=====]									
3.2 Search & Filters					[=====]								
3.3 Cart & Order Engine						[=====]							
3.4 Admin Panel UI							[=====]						
1.2													
PHASE 4: INTEGRATIONS													
4.1 Stripe Gateway Node								[=====]					
4.2 Shipping & Mail APIs									[=====]				
3.3													
PHASE 5: QA & DEPLOY													
5.1 End-to-End Testing											[=====]		
5.2 Cloud Deploy & CDN													
5.3 Production Launch													
5.1													
5.2													

2. Key Project Milestones

Milestones represent critical check-gates that require verification before entering subsequent phases:

- Milestone 1 (End of Week 2): Design Freeze.** Figma mockups are approved; frontend developers can safely code interfaces without facing design modifications.

- **Milestone 2 (End of Week 4): Backend Foundation Ready.** Relational schemas and authentication systems are live and verified.
- **Milestone 3 (End of Week 9): Feature Complete.** Core e-commerce mechanics (search, cart, check out, and integrations) are fully functional on local staging environments.
- **Milestone 4 (End of Week 12): Production Launch.** The platform passes security compliance filters and goes live to the public.

3. Managing Task Dependencies

To protect the launch timeline, project managers monitor the **critical path** created by task linkages:

- **Design to Code Integration:** The **Storefront Screens (3.1)** depend entirely on **Wireframes & Mockups (1.2)**. If design deliverables slip by 3 days, the frontend code start date slips uniformly.
- **Transactional Architecture Integration:** The **Stripe Gateway Node (4.1)** cannot collect payment values until the backend calculation logic in the **Cart & Order Engine (3.3)** is finalized.
- **Staging Validation Gate: End-to-End Testing (5.1)** cannot commence while external transactional structures are unlinked, as automated QA routines simulate a complete customer journey from search to final invoicing.

17.0 PROJECT TITLE

Title: Design and Development of an E-Commerce Web Application

Design & Development of a Modern E-Commerce Web Application

18.0 INTRODUCTION:

This project aims to design and develop a full-featured E-Commerce Web Application that enables users to browse products, place orders, make secure payments, and track deliveries. The system will also include an Admin Dashboard for managing products, orders, customers, and website content.

19.0 PROBLEM STATEMENT

Traditional retail limits customers by distance and availability, and small businesses lack affordable digital platforms. A secure, user-friendly, and scalable online shopping system is required.

20.0 OBJECTIVES

Main Objective:

To develop a fully functional and secure web-based e-commerce application.

Specific Objectives:

- Provide a user-friendly product browsing and purchasing interface.
- Implement secure authentication and payment processing.
- Build an admin panel for product, category, and order management.
- Ensure responsive design for different devices.
- Maintain secure data handling and storage.

21.0 SCOPE OF THE PROJECT

User Side:

- Registration & login
- Product browsing, search, and filters
- Cart & wishlist
- Checkout & payment

- Order tracking
- Ratings & reviews

Admin Side:

- User management
- Product CRUD operations
- Category & inventory management
- Order management
- Sales reports

22.0 PROPOSED SYSTEM OVERVIEW

Workflow:

1. User browses products.
2. Adds items to cart.
3. Checkout and payment.
4. Order placement.
5. Admin manages stock and orders.

23.0 TOOLS AND TECHNOLOGIES

Frontend:	HTML, CSS, JavaScript, React.js, Bootstrap/Tailwind
Backend:	Node.js/Express or PHP Laravel
Database:	MySQL or MongoDB
Authentication:	JWT/Sessions
Payment Gateway:	PayPal/Stripe/JazzCash
Version Control:	Git & GitHub

24.0 METHODOLOGY

Agile / SDLC approach:

- Requirement Analysis
- System Design (ERD, UML)
- UI/UX Design
- Frontend Development
- Backend Development
- Integration & Testing
- Deployment

25.0 EXPECTED OUTCOMES

- Fully functional e-commerce website
- Secure authentication & payment
- Admin dashboard
- Responsive UI
- Structured database
- Project documentation

26.0 PROJECT TIMELINE

Requirement Gathering	–	1 Week
Design –		1–2 Weeks
Frontend –		2 Weeks
Backend –		3 Weeks
Testing –		1 Week
Final Report –		1 Week
Total Duration:		8–10 Weeks

27.0 PROPOSAL CONCLUSION

The system will serve as a modern digital shopping platform with secure, responsive, and scalable features, suitable as a final year university project.

6. CONCLUSION

This project successfully demonstrates the design, deployment, and testing of a decoupled E-Commerce Web Application. By separating front-end interface components from back-end application layers, the system avoids the performance bottlenecks and single points of failure common in older web designs.

Rigorous integration testing confirmed that transaction pipelines handle concurrent checkout stresses safely while protecting critical user authentication profiles. The architecture provides a highly stable, extensible foundation ready for production-level retail environments.

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7. FUTURE SCOPE

Modern deployments must scale effectively alongside business growth. Future development paths for this application include:

- **AI Product Recommendations:** Integrating predictive engines to suggest relevant products based on user browsing history.
- **Microservices Evolution:** Splitting the core backend into individual services for payment processing, product catalogs, and order management.

Global Edge Caching: Storing static product images and asset libraries closer to international users to minimize site latency.

1. Vendor & B2B Management Module

The current server codebase entirely lacks specific vendor routing and authentication.

- **Vendor Portal:** Build dedicated registration, onboarding workflows, and restricted dashboard access (/api/vendor) to allow third-party sellers to isolate their business operations from general system administrators.
- **Supply Chain Automation:** Implement sequential workflows for **Purchase Order Creation, Sales Orders Management**, and automated **Sales/Purchase Invoices** matching the transactional pipelines outlined in your blueprint.

2. Financial Integration & Order Processing

The codebase currently relies on a localized /routes/orders.js file but lacks downstream enterprise services.

- **External Payment Gateway Integration:** Connect a third-party microservice (such as Stripe, PayPal, or Adyen API) into the backend server to securely execute remote authorizations, charge Captures, and structured **Refund Transactions**.
- **Real-time Logistics Engine:** Integrate dynamic third-party shipping APIs (like FedEx, UPS, or Shippo) inside repository.js to calculate spatial shipping rates dynamically at checkout rather than hardcoding static metrics.

3. Administrative Control & Security Enhancements

The codebase contains a generic middleware/auth.js helper but lacks explicit scoping controls.

- **Role-Based Access Control (RBAC):** Refactor the security middleware to distinctly gate administrative endpoints (e.g., locking down /api/products modification methods like POST/PUT/DELETE exclusively for Administrator or Vendor scopes).
- **Enterprise Reporting Engine:** Develop an analytics worker script capable of parsing sales logs to compile comprehensive, real-time **Sales/Purchase Reports** on demand.

4. Codebase Architecture & Technical Debt

- **Database Consolidation:** Decide whether to finalize the database infrastructure on a relational path (models-sql.js) or an unstructured document engine (models-mongo.js) to

eliminate redundant mapping layers in your repository pattern implementation.

- **Frontend Modernization:** Migrate the decoupled, raw vanilla script components (client/js/*.js) into a structured component-driven single-page framework (such as React or Vue.js) to streamline application state handling for features like shopping carts and multi-tiered filtering.

MUHAMMAD WASEEM AKHTAR, MUHAMMAD ALI DANISH & JASTY MARRY (BSCS FINAL YEAR PROJECT REPORT)

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